Analytical and Bioanalytical Chemistry (2026) 418:6891–6915
https://doi.org/10.1007/s00216-026-06687-9

RESEARCH PAPER

HILIC-MS/MS-based strategy for quantitative amino acid profiling in multiple biological matrices and its application in kidney disease samples

Xiongwei Yin[1] · Jean-Paul Decuypere[2] · Djalila Mekahli[2,3] · Bert Bammens[4,5] · Rudi Vennekens[6] · Erwin Adams[1] · Ann Van Schepdael[1]

Received: 3 June 2026 / Revised: 10 July 2026 / Accepted: 14 July 2026 / Published online: 4 August 2026
© The Author(s), under exclusive licence to Springer-Verlag GmbH, DE part of Springer Nature 2026

Abstract

Amino acid metabolism disorders have strong links to various kidney diseases, and the kidney is essential for maintaining systemic amino acid homeostasis. Given the lack of obvious clinical symptoms in the early stages of kidney disease, searching for reliable metabolic biomarkers is essential for early diagnosis. Endogenous AAs are widely present in human biological matrices, making it difficult to obtain analyte-free blank matrices for method validation. This poses challenges in the preparation of calibration standards, leading to uncertainty and reduced consistency in the results. This study aimed to develop a rapid hydrophilic interaction liquid chromatographic method coupled with tandem mass spectrometry. It could analyze 47 amino acids and related compounds, present in various biological matrices (human plasma, urine, and cyst fluids). The calibration curves were constructed by stripping each matrix to obtain a surrogate matrix. The slopes of the different calibration curves were systematically evaluated and the analytical results were compared with results obtained through the method of standard addition. The calibration curves established using both the standard addition method and the matrix stripping method were parallel, hence matrix stripping effectively mitigates matrix effects under a variety of matrix conditions, thereby assuring the accuracy and reliability of quantitative analysis results. To further investigate the applicability of this method to large-scale sample analysis, the analysis of plasma samples and urine samples obtained from patients with two types of kidney disease was performed.

Keywords Amino acids · LC-MS/MS · Plasma · Urine · Cyst fluid · Matrix effect

Introduction

Amino acids (AAs) are the fundamental building blocks of proteins and are essential for energy metabolism, signal transduction, and the maintenance of normal physiological functions [1]. During glomerular filtration and reabsorption in the proximal tubule, the kidneys maintain the body's dynamic homeostasis of AAs. Aside from being extensively involved in AA synthesis, transport, and metabolic regulation, the kidney also ensures their efficient utilization under various physiological conditions [2, 3]. A variety of kidney conditions have been linked to changes in AA concentrations, which are closely related to renal dysfunction [4]. Researchers have found that abnormal changes in AA profiles are associated with various kidney diseases, such as chronic kidney disease (CKD) [5], acute kidney injury [6], diabetic kidney disease [7], and autosomal dominant polycystic kidney disease (ADPKD) [8]. As a result, these

✉ Ann Van Schepdael
ann.vanschepdael@kuleuven.be

1 Department of Pharmaceutical and Pharmacological Sciences, Pharmaceutical Analysis, KU Leuven, O&N2, Herestraat 49 Bus 923, 3000 Leuven, Belgium

2 PKD Research Group, Department of Cellular and Molecular Medicine, KU Leuven, Leuven, Belgium

3 Department of Pediatric Nephrology, University Hospitals Leuven, Leuven, Belgium

4 Department of Nephrology, University Hospitals Leuven, Leuven, Belgium

5 Nephrology and Renal Transplantation Research Group, Department of Microbiology, Immunology and Transplantation, KU Leuven, Leuven, Belgium

6 Laboratory of Ion Channel Research, Department of Cellular and Molecular Medicine, KU Leuven, Leuven, Belgium

changes occur because of pathological processes that may involve impairment of glomerular filtration function, tubular reabsorption disorders, or disturbances of metabolic pathways. Since kidney disease is often characterized by the absence of obvious clinical symptoms at the beginning of its course, identifying reliable metabolic biomarkers is crucial for early detection [9, 10]. It is likely that amino acid metabolism disorders will not only serve as indicators of kidney function but will also provide important clues to elucidate the molecular mechanisms that contribute to kidney disease and to create potential therapeutic targets.

Among the branches of metabolomics, targeted amino acidomics occupies a prominent position. In addition to serving as highly valuable target molecules, AAs can reveal the molecular connection between AA metabolism and the development and onset of diseases, providing an opportunity to gain a deeper understanding of disease pathogenesis and identify potential biomarkers for clinical diagnosis and personalized treatment [11, 12]. It can be applied to the study of a wide range of biological fluid samples, including plasma, urine, cerebrospinal fluid, and cyst fluid. Through the construction of metabolic pathways and analysis of metabolite networks, it is possible to obtain a systematic reflection of metabolic status and associated pathophysiological changes within the body [13, 14]. As AAs and their related metabolites are typically highly polar, reverse-phase liquid chromatography (RPLC) is not a suitable technique for their separation. Derivatization by chemical means can improve the chromatographic behavior and detection sensitivity of a compound. Still, it is cumbersome, prone to generating byproducts, and may adversely impact its repeatability and accuracy [15]. On the other hand, hydrophilic interaction liquid chromatography (HILIC) exhibits greater separation efficiency and stability when used in AA analysis. This technique is not only free of derivatization requirements, but it also improves the spraying and ionization efficiency in MS by utilizing a high ratio of organic solvent to water in the mobile phase [16]. Because of the continuous advancement of MS technology, LC-triple quadrupole MS has become the main method for quantifying AAs using multiple reaction monitoring (MRM) due to its high efficiency, high sensitivity, and high selectivity. The technique is widely used in both basic research and clinical translational research [17].

Quantitative analysis of endogenous compounds is a key tool in clinical diagnosis and monitoring, drug metabolism research, and nutritional assessment. It can provide important information on the physiological state of the organism and facilitate the early discovery of biomarkers. However, due to the complexity of biological samples and the diversity of endogenous compounds, quantitative analysis faces many challenges [18]. It is often necessary to consider factors such as the complexity of the compounds in the sample, large differences in concentration ranges, and matrix interference [19]. In terms of the analytical methods for endogenous compounds, the FDA clearly states in its method validation guidelines that the best option for evaluating an analytical method is to add the target analyte to the real matrix [20]. However, a true blank matrix is usually difficult to obtain in the case of endogenous compounds, as it is challenging to completely remove the analyte while retaining the original characteristics of the matrix. This limitation can affect the assessment of matrix effects and recovery rates, which in turn leads to inaccurate quantitative results during method validation [21]. Quantitative detection of endogenous compounds has always been a major challenge in bioanalysis. Of the four common analytical methods, the most accurate is the standard addition method, in which a standard substance in several known concentrations is added to the sample, a calibration curve is plotted, and extrapolated to the abscissa to calculate the concentration of the endogenous compound in the original sample [22]. However, a large volume of samples is required, and the process is time-consuming, making it unsuitable for the quantitative analysis of large sample sizes in the clinic. Another method is background subtraction. This method has the disadvantages of different levels of endogenous substances in blank matrix batches, batch-to-batch differences, poor reproducibility, and difficulty in comparing results from multiple laboratories horizontally [23]. Besides, interfering substances in complex biological matrices may overlap with the signals of endogenous compounds, making it difficult to accurately deduce the background signal and affecting the accuracy of quantification. In addition to the above two methods, the two methods that are used more frequently are the surrogate analyte and surrogate matrix. The introduction of isotope-labeled internal standards (ISs) as surrogate analytes is a common method because isotopes are similar in chemical properties to the target analytes and are less affected by endogenous backgrounds [24]. However, not all substances have corresponding ISs, and the use of several ISs as standards at the same time further increases costs, making this method expensive. The most used surrogate matrix method includes the use of pure solutions based on neat solvents, artificial biological fluid matrices, and stripped matrices [25]. Although pure solutions and artificial matrices are commonly used, they have limitations when replacing real matrices, especially in terms of matrix effects and recovery rate assessment [26]. The advantage of stripped matrices is that by removing endogenous compounds, they reduce matrix interference with the analysis, thereby significantly improving the sensitivity of the detection and the accuracy of the data.

This work aimed at establishing a HILIC-MS/MS analytical approach that enables the detection of 47 AAs and associated compounds across various biological matrices, for example, plasma, urine, and cyst fluids from humans. The stripped matrix method was utilized to prepare blank

matrices for constructing calibration curves. The slopes of the different calibration curves were then systematically compared, in comparison with the standard addition method, fully demonstrating the accuracy and reliability of the method. Method validation was conducted for the target analytes and comprised the following items: linearity, limit of quantification, precision, accuracy, recovery, and sample stability. To further validate the applicability of this method for large-scale investigations, analyses were conducted on plasma, urine, and cyst fluid samples obtained from patients with two kidney diseases.

Materials and methods

Chemicals and reagents

In this study, 47 AA standards were utilized, of which 42 were obtained from Sigma-Aldrich and 5 were obtained from other manufacturers: 3-cyclohexyl-d-alanine hydrate (Fluka Analytical, Bucharest, Romania), l-cysteine (Federa SC SV, Brussels, Belgium), N-acetyl-l-cysteine (Acros Organics, Ghent, Belgium), (R)-(-)−2-(2,5-dihydroxyphenyl)glycine and l-(+)-α-phenylglycine (Janssen, Geel, Belgium). Hydrochloric acid (HCl, 37%, Fisher Scientific, Merelbeke, Belgium), methanol (MeOH, LC-MS grade, Fisher Scientific, Loughborough, UK), and water (LC-MS grade, Fisher Scientific, Loughborough, UK) were all purchased from Fisher Scientific. Acetonitrile (LC-MS grade) was purchased from Biosolve Chimie SARL (Dieppe, France). Sigma-Aldrich company provided a mixture of 20 ^{13}C and ^{15}N isotope-labeled proteinogenic AAs, ammonium formate, formic acid, and activated carbon. 0.22 µm PTFE filters were purchased from BGB Analytics (Haarlem, the Netherlands).

Human plasma, urine, and cyst fluid samples

A total of 39 plasma samples (EDTA-plasma) and urine samples from patients have been collected. Patients with the two kidney diseases included those from stage 1 to stage 4, comprising 19 cases of CKD and 20 cases of ADPKD. Additionally, one cyst fluid sample has been collected from an ADPKD patient. All samples were obtained from a university hospital (study approved by the Ethics Committee Research of UZ Leuven under number S50495).

Standard preparation

L-asparagine, *N*-acetyl-L-cysteine, L-(+)-α-phenylglycine, (R)-(-)−2-(2,5-dihydroxyphenyl)glycine, L-norvaline, L-methionine, 3-nitro-L-tyrosine, and L-glutamine were dissolved in 0.1 M HCl to prepare a 10 mM solution separately; kynurenine and 3-cyclohexyl-D-alanine were dissolved in 0.1 M HCl to prepare a 5 mM solution; glycine and L-taurine were dissolved in 1 M HCl to prepare a 5 mM solution; the remaining 35 AAs were dissolved in 0.1 M HCl to prepare a 20 mM solution.

Sample preparation

Plasma samples were processed according to the following procedure: mix 50 µL of plasma with an equal volume of water, add 296 µL of methanol-acetonitrile solution containing 0.1% formic acid (volume ratio 1:3) (protein precipitation solution), and precipitate the protein using 4 µL isotope-labeled IS mixture. In the following step, the mixture was vortexed for 30 s, centrifuged at 14,100 g for 15 min, and then filtered through a 0.22 µm PTFE membrane filter before analysis. The procedures followed for preparing urine samples were similar to those used for preparing plasma samples. Urine samples were first diluted with water at a ratio of 1:2 and then processed according to the same procedures established for plasma samples. The cyst fluid sample should be handled in the following way: 25 µL of sample should be mixed with 25 µL of water. Next, 164 µL protein precipitation solution should be added, as well as 4 µL isotope IS solution. The next processing of the samples followed the same steps as those for plasma samples.

Stripped blank matrix

The blank plasma, urine, and cyst fluids' matrix were all prepared using the same procedure, following a two-step centrifugation method [27]. In this experiment, plasma, urine, and cyst fluids were each mixed at a ratio of one milliliter of supernatant from the first centrifugation and 100 mg of activated carbon. After shaking the mixture for 8 h at 4 °C and 1400 rpm in a Thermomix Comfort shaker (Eppendorf, Aarschot, Belgium), the matrix was centrifuged (Minispin Plus centrifuge (Eppendorf)) at 14,100 g for 30 min. The sediment was decanted, and the supernatant was collected and stored at −20 °C.

LC-MS/MS analysis

All samples were analyzed on an Agilent 1100 HPLC system equipped with a binary high-pressure gradient pump, a column oven (G1316A), and an autosampler (G1367A) (Agilent Technologies, Santa Clara, CA, USA). Separation employed an XBridge BEH Amide column (2.1 × 150 mm, 3.5 µm, 130 Å) with a guard column (2.1 × 5 mm) (Waters, Milford, MA, USA). Mobile phase A was 10 mM ammonium formate with 0.15% formic acid in 5% acetonitrile, while mobile phase B was the same solution in 95% acetonitrile. Gradient elution and ESI positive ion mode conditions are detailed in Table 1.

Table 1 MS parameters of AAs, their related compounds, and corresponding ISs

	AAs	ISs	DP (volts)	FP (volts)	EP (volts)	Precursor ion *(m/z)*	Product ion *(m/z)*	CE (volts)	CXP (volts)	RT (min)
1	*N*-acetyl-l-cysteine	L-cystine ^{13}C ^{15}N	19	195	5	164	122[a]	11	6	1.56
							76[b]	26	6	
2	3-cyclohexyl-D-alanine hydrate	L-alanine ^{13}C ^{15}N	34	200	9	172.3	126.3[a]	18	8	4.62
							109.1[b]	25	8	
							83.1[b]	29	12	
3	3-nitro-l-tyrosine	L-tyrosine ^{13}C ^{15}N	30	190	9	227	181.1[a]	19	10	4.62
							168.1[b]	25	8	
	L-phenylalanine ^{13}C ^{15}N		32	225	6	176.2	129.2[a]	17	6	4.65
4	L-phenylalanine	L-phenylalanine ^{13}C ^{15}N	32	225	6	166.1	120.2[a]	17	6	4.67
							103.1[b]	35	8	
	L-tryptophan ^{13}C ^{15}N		52	215	11	217.9	200[a]	14	12	4.69
5	L-tryptophan	L-tryptophan ^{13}C ^{15}N	52	215	11	205.2	188[a]	14	12	4.72
							146.2[b]	23	8	
6	L-5-oxoproline	L-glutamic acid ^{13}C ^{15}N	48	240	9	130.1	84.1[a]	18	14	4.72
							56.1[b]	37	8	
7	kynurenine	L-tryptophan ^{13}C ^{15}N	32	234	11	209.1	192.1[a]	12	12	4.79
							94.1[b]	22	4	
8	L-leucine	L-Leucine ^{13}C ^{15}N	20	193	9	132.2	43.2[a]	37	7	4.86
							86.1[b]	16	7	
9	L-isoleucine	L-Leucine ^{13}C ^{15}N	29	234	7	132.1	69[a]	23	6	5.27
							86.2[b]	16	15	
	L-leucine ^{13}C ^{15}N		20	193	9	139.1	92[a]	16	7	4.87
10	L-(+)-alpha-phenylglycine	L-glycine ^{13}C ^{15}N	17	199	8	152.2	135[a]	14	7	5.38
11	ethanolamine	L-serine ^{13}C ^{15}N	20	150	7	62.1	44.1[a]	14	7	5.41
							45[b]	15	6	
							62[b]	5	11	
12	(R)-(-)−2-(2,5-dihydrophenyl)glycine	L-glycine ^{13}C ^{15}N	31	265	10	154.2	91.1[a]	22	4	5.48
							137.1[b]	12	8	
	L-methionine ^{13}C ^{15}N		55	192	11	155.8	109.1[a]	14	9	5.76
13	L-methionine	L-methionine ^{13}C ^{15}N	55	192	11	150.2	104.1[a]	14	9	5.77
							133[b]	13	7	
14	norvaline	L-valine ^{13}C ^{15}N	21	220	10	117.9	72.3[a]	13	13	6.07
							55.1[b]	32	4	
15	γ-aminobutyric acid	L-glutamic acid ^{13}C ^{15}N	33	129	7	104.1	87.2[a]	15	7	6.13
							69.1[b]	21	12	
16	L-valine	L-valine ^{13}C ^{15}N	27	153	7	118.2	72.2[a]	15	13	6.55
							55.1[b]	29	9	
	L-valine ^{13}C ^{15}N		27	153	7	123.9	77.1[a]	15	13	6.54
17	L-proline	L-proline ^{13}C ^{15}N	37	227	7	116.2	70.2[a]	21	12	6.79
							43.2[b]	34	10	
	L-proline ^{13}C ^{15}N		37	227	7	122	75.1[a]	21	12	6.78
18	DL-β-aminoisobutyric acid	L-glutamic acid ^{13}C ^{15}N	30	250	10	104	41.3[a]	37	6	6.81
							58.1[b]	14	4	
19	L-tyrosine	L-tyrosine ^{13}C ^{15}N	32	194	9	182.1	136.2[a]	17	8	6.82
							123.2[b]	27	10	
	L-tyrosine ^{13}C ^{15}N		32	194	9	191.9	145[a]	17	8	6.84
20	L-taurine	L-cystine ^{13}C ^{15}N	52	211	11	126.1	107.9[a]	15	6	7.11
							126.1[b]	8	5	

Table 1 (continued)

	AAs	ISs	DP (volts)	FP (volts)	EP (volts)	Precursor ion *(m/z)*	Product ion *(m/z)*	CE (volts)	CXP (volts)	RT (min)
21	L-cysteine	L-cystine ^{13}C ^{15}N	28	175	9	122.1	76.1[a]	18	14	7.50
							59.1[b]	32	5	
22	methionine sulfone	L-methionine ^{13}C ^{15}N	20	200	8	182.1	56[a]	15	16	7.59
							136[b]	5	6	
23	L-α-amino-n-butyric acid	L-valine ^{13}C ^{15}N	29	197	6	104.1	58.1[a]	14	9	7.66
							41.2[b]	40	6	
24	sarcosine	L-alanine ^{13}C ^{15}N	28	212	7	90.2	44.1[a]	16	7	8.02
							72[b]	6	12	
25	β-alanine	L-alanine ^{13}C ^{15}N	25	159	7	90.1	72[a]	11	12	8.35
							30.1[b]	27	4	
	L-alanine ^{13}C ^{15}N		40	175	10	94.1	47.1[a]	16	7	8.65
26	L-alanine	L-alanine ^{13}C ^{15}N	40	175	10	90.1	44.2[a]	16	7	8.67
							62.1[b]	9	12	
27	*trans*−4-hydroxy-L-proline	L-Leucine ^{13}C ^{15}N	26	158	10	132.1	86.1[a]	18	7	8.76
							68.2[b]	28	6	
	L-threonine ^{13}C ^{15}N		38	141	6	124.8	78[a]	16	12	9.28
28	L-threonine	L-threonine 13 C 15N	38	141	6	120.2	74.2[a]	16	12	9.30
							102[b]	12	16	
	L-glycine ^{13}C ^{15}N		31	172	5	79.1	32.1[a]	17	11	9.58
29	L-glycine	L-glycine ^{13}C ^{15}N	31	172	5	76	30.1[a]	17	11	9.60
							48.1[b]	7	7	
							130.2[b]	14	8	
30	L-glutamine	L-glutamine ^{13}C ^{15}N	20	245	9	147.1	84[a]	25	4	10.39
							130.2[b]	14	8	
	L-glutamine ^{13}C ^{15}N		20	245	9	153.9	89[a]	25	4	10.42
	L-serine ^{13}C ^{15}N		30	123	10	110.1	92.2[a]	13	4	10.43
31	L-serine	L-serine ^{13}C ^{15}N	30	123	10	106.1	88.1[a]	13	4	10.45
							60.1[b]	15	11	
	L-asparagine ^{13}C ^{15}N		27	205	9	139	76.9[a]	22	13	10.64
32	L-asparagine	L-asparagine ^{13}C ^{15}N	27	205	9	133.1	74[a]	22	13	10.67
							87.1[b]	14	4	
	L-glutamic acid ^{13}C ^{15}N		28	193	9	154.3	136.3[a]	13	7	10.90
33	L-glutamic acid	L-glutamic acid ^{13}C ^{15}N	28	193	9	148.2	130.1[a]	13	7	10.93
							84.1[b]	24	15	
34	L-citrulline	L-arginine ^{13}C ^{15}N	31	177	8	176.1	159.1[a]	14	9	11.24
							70[b]	36	13	
35	1-methyl-l-histidine	L-histidine ^{13}C ^{15}N	29	220	11	170.1	124.1[a]	27	7	11.59
							83[b]	28	14	
							96.1[b]	30	5	
	L-aspartic acid ^{13}C ^{15}N		29	157	8	139.1	77[a]	8	14	11.99
36	L-aspartic acid	L-aspartic acid ^{13}C ^{15}N	29	157	8	134.1	74.1[a]	14	17	12.02
							88.1[b]	8	14	
37	L-histidine	L-histidine ^{13}C ^{15}N	30	150	8	156.1	110.1[a]	20	6	12.61
							83.1[b]	38	6	
	L-histidine ^{13}C ^{15}N		30	150	8	165.3	118.1[a]	20	6	12.61
38	3-methyl-l-histidine	L-histidine ^{13}C ^{15}N	45	200	10	170	96.2[a]	23	5	12.83
							126.2[b]	19	7	
							109.1[b]	23	6	

Table 1 (continued)

	AAs	ISs	DP (volts)	FP (volts)	EP (volts)	Precursor ion (*m/z*)	Product ion (*m/z*)	CE (volts)	CXP (volts)	RT (min)
	L-arginine ^{13}C ^{15}N		36	155	11	184.7	75.1[a]	29	12	12.91
39	L-arginine	L-arginine ^{13}C ^{15}N	36	155	11	175.2	70.2[a]	29	12	12.92
							116.1[b]	20	6	
	L-lysine ^{13}C ^{15}N		25	177	9	155.2	90.1[a]	23	8	13.31
40	L-lysine	L-lysine ^{13}C ^{15}N	25	177	9	147.3	84.2[a]	23	8	13.33
							130.1[b]	13	7	
41	homocystine	L-methionine ^{13}C ^{15}N	23	200	11	269.1	136[a]	14	8	13.37
							88.2[b]	34	8	
42	L-ornithine	L-arginine ^{13}C ^{15}N	31	214	7	133.2	70.2[a]	24	12	13.41
							116.1[b]	13	6	
43	anserine	L-histidine ^{13}C ^{15}N	23	209	9	241.3	109.1[a]	36	6	13.43
44	carnosine	L-histidine ^{13}C ^{15}N	29	210	8	227.1	210[a]	16	13	13.47
							110.2[b]	34	5	
45	δ-hydroxylysine	L-lysine ^{13}C ^{15}N	20	180	11	163	128.2[a]	17	9	14.09
							82[b]	23	12	
46	cystathionine	L-serine ^{13}C ^{15}N	35	185	11	223	88.1[a]	51	8	14.33
							134[b]	24	10	
	L-cystine ^{13}C ^{15}N		31	238	9	249	156.1[a]	18	9	14.49
47	L-cystine	L-cystine ^{13}C ^{15}N	31	238	9	241.1	152.1[a]	18	9	14.50
							74.1[b]	43	13	

DP declustering potential, *FP* focusing potential, *EP* entrance potential, *CE* collision energy, *CXP* collision cell exit potential, *RT* retention time
[a]Ion selected for quantification, [b]ion selected for confirmation

Method validation

European Medicines Agency (EMA) guidelines and recommendations regarding bioanalytical methods were referenced in the method validation process [28].

Linearity of the calibration curve and sensitivity

Limit of detection (LOD) can be defined as signal to noise ratio (S/N) ≥ 3 and limit of quantification (LOQ) as S/N ≥ 10. In calibration curves containing at least six concentrations, linear regression (weighting factor of 1/x) of analyte-to-IS peak area ratios versus concentration was used.

Accuracy and precision

The precision and accuracy for the different matrices were tested at three concentrations, low quality control, medium quality control, and high quality control (LQC, MQC, and HQC). The LQC was considered at nearly threefold the LOQ, whereas the MQC and HQC were formulated within the linearity range, at about 50% and 85% of that range, respectively. Each concentration was analyzed six times within a single day to determine the intra-day precision, whereas the analysis of each concentration over three consecutive days determined the inter-day precision ($n = 6$).

Matrix effects

Matrix effects were evaluated by comparing the slopes of the calibration curves. An analysis of the slopes of calibration curves for the analyte and the IS compounds was conducted using real matrix, stripped matrix, and pure water, respectively ($n = 3$). The differences in slopes between different systems reflect the influence of matrix components on the analytical response. This also facilitates the use of the standard addition method to extrapolate the concentrations of endogenous compounds in the real matrix and compare them with those obtained in the stripped matrix and pure water.

Carry-over

The carry-over is evaluated by assessing the blank stripped sample after three consecutive HQC injections. Blank samples must not have residual analyte content over 20% and ISs cannot have residual content over 5%.

Recovery

The recovery is assessed by comparing the signal responses before and after extraction: the peak areas obtained after spiking the analyte at low, medium, and high concentration levels into the stripped matrix and completing extraction are compared with the peak areas obtained after adding the analyte after extraction. Six parallel samples are prepared for each level.

Stability

Two conditions were tested for stability using three concentrations and six parallel samples. Storage at 4 °C for 24 h simulates short-term autosampler conditions, while three freeze-thaw cycles assess the stability during repeated thawing and reanalysis. For testing, three cycles of freezing and thawing were performed on the samples after they had been kept at 4 °C for 24 h.

Statistical analysis

The differential distribution of AAs and their derivatives between patients with ADPKD and those with CKD was visualized using orthogonal partial least squares discriminant analysis (OPLS-DA) in MetaboAnalyst 6.0 (https://www.metaboanalyst.ca/). Statistical analyses were performed using SPSS version 27 (IBM) and GraphPad Prism version 10.1.2 software. To compare the two patient groups, the Mann–Whitney U test and the unpaired two-tailed Student's t test were applied. AAs and their derivatives showing significant differences were identified based on the criteria of Variable Importance in Projection (VIP) > 1 and $p < 0.05$.

Results and discussion

Method validation

The results of linearity of the calibration curve, precision, accuracy, LOD, LOQ, and recovery for urine and cystic fluid samples are described below, while the plasma results are described in [27].

Linearity of the calibration curve and carry-over

Based on Tables 2 and 3, the calibration curves for all target compounds showed excellent linearity, with $R^2 > 0.99$. The residuals were randomly distributed. Carry-over was negligible for all target compounds and did not interfere with quantification.

Accuracy and precision

Table 4 shows that in the urine matrix, the intra-day accuracy ranged from 87.4% to 108.6% for LQC, 89.8% to 111.6% for MQC, and 87.4% to 113.5% for HQC. The corresponding RSD values were 3.3%–11.7%, 2.5%−11.6%, and 3.0%–9.2%, respectively. For inter-day analysis, the accuracy was 86.5%–112.4% for LQC, 88.6%–114.3% for MQC, and 87.1%–112.8% for HQC, with RSD values ranging from 3.9%–13.9%, 1.8%–13.2%, and 2.5%–11.8%, respectively. Precision and accuracy results demonstrated that intra- and inter-day variations remained within acceptable bioanalytical standards, confirming the reliability of the method.

Table 5 shows that in the cyst fluids, intra-day accuracy was 88.3%–110.0% for LQC, 86.4%–101.3% for MQC, and 88.0%–111.9% for HQC, with RSD values of 2.5%–14.5%, 1.4%−14.5%, and 2.2%–12.1%, respectively. Inter-day assessments showed accuracies of 86.5%–113.1% (LQC), 85.4%-−110.0% (MQC), and 89.1%–112.2% (HQC), with RSDs ranging from 5.1%–13.0%, 2.1%–14.5%, and 1.7%–14.3%, respectively. The accuracy and precision values met the method validation requirements, indicating that the method provides accurate measurements, ensuring reliable quantification results across the linearity range.

Matrix effects

During this study, matrix effects were evaluated through comparison of the slopes of calibration curves obtained for the analytes for both the surrogate matrix and the real matrix. It means the slope ratio is obtained by dividing the slope of the surrogate matrix calibration curve by the slope of the real matrix calibration curve. Three types of matrices were investigated in this study, including plasma (see [27] for the underlying data), urine, and cyst fluids. Based on the results (see Electronic Supplementary Material Figures S1 and S2 and Fig. 1), slope ratios range between 0.8 and 1.2 for most compounds in the stripped matrices [26, 29], suggesting that the influence of the matrix effect on the analytical result is properly mitigated using the stripped matrix. On the other hand, when water was used as an alternative matrix, a significant matrix effect was observed in approximately one-third to nearly half of the compounds.

Further comparing the concentrations of analytes obtained by the standard addition method with those obtained by the surrogate matrix method revealed that the results obtained by the stripped matrix method were closer to those obtained by the standard addition method, in comparison with water-based calibration curves. EMA bioanalytical requirements state that all concentration levels must

Table 2 Linear range, LOQ, and LOD of AAs and their related compounds in stripped urine

AAs	LOD (μM)	LOQ (μM)	Linearity (μM)	R^2
N-acetyl-l-cysteine	0.35	0.87	1.73–222	0.999
3-cyclohexyl-D-alanine hydrate	0.11	0.44	0.87–111	0.994
3-nitro-l-tyrosine	0.44	0.87	1.73–222	0.995
L-phenylalanine	0.20	0.48	1.95–250	0.998
L-tryptophan	0.49	0.98	1.95–250	0.998
L-5-oxoproline	0.75	1.08	17.34–2220	0.997
kynurenine	0.21	0.43	0.86–111	0.994
L-leucine	0.39	0.65	1.95–250	0.992
L-isoleucine	0.49	0.98	1.95–250	0.997
L-(+)-alpha-phenylglycine	0.27	0.87	1.73–222	0.999
ethanolamine	0.29	0.98	1.95–250	0.993
(R)-(-)−2-(2,5-dihydrophenyl)glycine	0.58	0.87	1.73–222	0.999
L-methionine	0.29	0.65	1.95–250	0.997
norvaline	0.22	0.87	1.73–222	0.991
γ-aminobutyric acid	0.65	0.98	1.95–250	0.999
L-valine	0.25	0.49	1.95–250	0.998
L-proline	0.29	0.49	1.95–250	0.995
DL-β-aminoisobutyric acid	0.98	1.95	3.91–250	0.996
L-tyrosine	0.25	0.49	1.95–250	0.999
L-taurine	0.35	0.87	1.73–222	0.999
L-cysteine	3.48	8.67	17.34–2220	0.995
methionine sulfone	0.44	0.87	1.73–222	0.994
L-α-amino-n-butyric acid	0.39	0.98	1.95–250	0.993
sarcosine	0.33	0.65	1.95–250	0.996
β-alanine	0.49	0.98	1.95–250	0.998
L-alanine	0.24	0.65	1.95–250	0.995
trans−4-hydroxy-L-proline	0.24	0.65	1.95–250	0.991
L-threonine	0.25	0.49	1.95–250	0.999
L-glycine	21.68	43.35	86.7–11,100	0.998
L-glutamine	0.35	0.87	1.73–222	0.993
L-serine	0.25	0.98	1.95–250	0.996
L-asparagine	0.35	0.87	1.73–222	0.997
L-glutamic acid	0.25	0.49	1.95–250	0.999
L-citrulline	0.25	0.49	1.95–250	0.992
1-methyl-l-histidine	0.12	0.24	1.95–250	0.998
L-aspartic acid	0.24	0.65	1.95–250	0.994
L-histidine	0.25	0.49	1.95–250	0.997
3-methyl-l-histidine	0.33	0.65	1.95–250	0.998
L-arginine	0.12	0.24	1.95–250	0.999
L-lysine	0.25	0.49	1.95–250	0.998
homocystine	0.49	0.98	1.95–250	0.999
L-ornithine	0.24	0.65	1.95–250	0.999
anserine	0.25	0.49	1.95–250	0.996
carnosine	0.20	0.49	1.95–250	0.991
δ-hydroxylysine	0.25	0.98	1.95–250	0.996
cystathionine	0.39	0.65	1.95–250	0.997
L-cystine	0.25	0.49	1.95–250	0.998

Table 3 Linear range, LOQ, and LOD of AAs and their related compounds in stripped cyst fluids

AAs	LOD (μM)	LOQ (μM)	Linearity (μM)	R^2
N-acetyl-l-cysteine	1.73	3.47	6.94–250	0.999
3-cyclohexyl-D-alanine hydrate	0.09	0.44	0.87–111	0.999
3-nitro-l-tyrosine	0.94	3.47	6.94–222	0.990
L-phenylalanine	0.65	0.98	1.95–250	0.999
L-tryptophan	0.49	0.98	1.95–250	0.994
L-5-oxoproline	4.34	8.67	17.34–2220	0.994
kynurenine	0.44	0.87	0.87–111	0.999
L-leucine	0.98	1.95	1.95–250	0.996
L-isoleucine	3.90	7.81	7.81–250	0.997
L-(+)-alpha-phenylglycine	0.87	1.73	1.73–222	0.998
ethanolamine	1.98	3.95	7.81–250	0.999
(R)-(-)−2-(2,5-dihydrophenyl)glycine	0.87	1.73	1.73–222	0.998
L-methionine	1.95	7.81	7.81–250	0.998
norvaline	0.44	0.87	1.73–222	0.991
γ-aminobutyric acid	0.13	0.25	1.95–250	0.999
L-valine	0.13	0.49	1.95–250	0.998
L-proline	0.87	1.95	3.90–250	0.996
DL-β-aminoisobutyric acid	1.95	15.63	15.63–250	0.999
L-tyrosine	0.98	3.90	3.90–250	0.998
L-taurine	1.73	6.94	13.88–222	0.999
L-cysteine	34.69	69.38	138.75–2220	0.999
methionine sulfone	6.94	13.88	13.88–222	0.995
L-α-amino-n-butyric acid	0.25	0.49	1.95–250	0.991
sarcosine	1.95	3.90	3.9–250	0.995
β-alanine	3.90	7.81	7.81–250	0.994
L-alanine	0.98	1.95	7.81–250	0.993
trans−4-hydroxy-L-proline	0.25	0.98	3.90–250	0.998
L-threonine	0.49	1.95	7.81–250	0.999
L-glycine	86.72	346.89	346.89–11,100	0.998
L-glutamine	0.87	1.73	3.47–222	0.998
L-serine	0.49	0.98	3.90–250	0.995
L-asparagine	0.43	1.73	3.47–222	0.997
L-glutamic acid	0.49	1.95	7.81–250	0.999
L-citrulline	0.49	1.95	7.81–250	0.994
1-methyl-l-histidine	0.25	0.98	3.90–250	0.993
L-aspartic acid	1.95	3.91	7.81–250	0.997
L-histidine	0.25	1.95	7.81–250	0.998
3-methyl-l-histidine	0.98	1.95	3.90–250	0.994
L-arginine	0.25	0.49	1.95–250	0.994
L-lysine	0.49	0.98	1.95–250	0.991
homocystine	0.65	1.98	7.81–250	0.995
L-ornithine	0.25	0.98	3.90–250	0.992
anserine	0.49	0.98	3.90–250	0.996
carnosine	0.49	1.95	7.81–250	0.995
δ-hydroxylysine	0.89	3.90	15.63–250	0.995
cystathionine	0.49	1.95	7.81–250	0.993
L-cystine	0.25	0.98	7.81–250	0.996

Table 4 Intra-day and Inter-day precision and accuracy of the analyzed AAs and their related compounds in stripped urine ($n=6$)

AAs		Intra-day		Inter-day			Intra-day		Inter-day			Intra-day		Inter-day	
		Accuracy (%)	Precision (RSD, %)	Accuracy (%)	Precision (RSD,%)		Accuracy (%)	Precision (RSD, %)	Accuracy (%)	Precision (RSD, %)		Accuracy (%)	Precision (RSD%)	Accuracy (%)	Precision (RSD, %)
N-acetyl-l-cysteine	LQC	92.0	4.6	108.5	7.6	MQC	101.7	6.0	93.5	6.9	HQC	98.6	4.1	99.2	4.4
3-cyclohexyl-D-alanine hydrate	LQC	101.1	7.7	99.1	11.9	MQC	103.8	5.3	93.8	13.2	HQC	100.0	4.0	95.3	10.7
3-nitro-L-tyrosine	LQC	100.0	7.9	91.8	11.4	MQC	89.1	2.6	89.3	7.6	HQC	90.8	3.3	95.7	8.7
L-phenylalanine	LQC	106.1	5.1	102.3	13.9	MQC	102.9	4.3	98.1	4.3	HQC	99.2	3.9	96.8	8.3
L-tryptophan	LQC	95.3	8.2	94.0	13.3	MQC	98.6	4.2	94.9	5.0	HQC	97.4	4.1	94.4	8.0
L-5-oxoproline	LQC	92.9	6.4	94.3	7.9	MQC	96.7	2.5	98.5	8.9	HQC	98.1	4.5	99.1	5.4
kynurenine	LQC	108.6	8.4	108.1	9.6	MQC	109.3	5.5	114.3	5.7	HQC	113.5	3.2	105.1	5.0
L-leucine	LQC	97.8	5.5	103.4	10.0	MQC	98.4	6.5	93.1	7.7	HQC	100.3	5.2	106.7	8.5
L-isoleucine	LQC	87.9	6.4	94.2	12.8	MQC	94.1	5.4	93.6	8.7	HQC	98.8	7.9	97.1	8.0
L-(+)-alpha-phenylglycine	LQC	92.8	8.1	104.0	6.4	MQC	95.7	5.9	98.5	6.3	HQC	97.2	5.2	98.8	8.5
ethanolamine	LQC	99.4	4.2	91.1	11.2	MQC	98.0	5.1	104.4	7.6	HQC	95.1	3.8	107.3	5.1
(R)-(-)−2-(2,5-dihydrophenyl) glycine	LQC	94.1	10.2	103.2	11.7	MQC	101.6	4.6	95.8	8.5	HQC	100.7	4.2	101.8	5.2
L-methionine	LQC	90.9	8.7	92.4	7.4	MQC	89.8	7.5	99.5	2.0	HQC	96.0	5.1	103.4	6.6
norvaline	LQC	99.3	5.0	102.1	9.0	MQC	90.3	3.8	93.5	9.1	HQC	95.4	4.1	95.6	5.2
γ-aminobutyric acid	LQC	87.4	10.5	108.7	11.9	MQC	98.0	7.4	88.6	6.2	HQC	103.6	5.6	99.1	6.0
L-valine	LQC	93.0	5.6	96.3	4.3	MQC	100.7	7.4	92.7	10.6	HQC	103.1	4.9	99.1	7.3
L-proline	LQC	97.6	3.3	95.8	4.6	MQC	99.9	5.9	94.7	8.3	HQC	96.1	3.4	88.5	3.5
DL-β-aminoisobutyric acid	LQC	92.1	10.1	93.4	9.4	MQC	101.0	10.0	95.2	9.4	HQC	99.9	9.2	100.0	3.6
L-tyrosine	LQC	103.3	11.3	102.5	5.2	MQC	93.8	11.6	92.4	11.5	HQC	87.5	8.3	89.5	3.5
L-taurine	LQC	99.4	8.7	93.9	11.8	MQC	91.6	8.9	91.0	4.3	HQC	105.2	6.5	97.9	9.1
L-cysteine	LQC	97.2	7.3	102.7	9.3	MQC	102.4	8.3	94.1	7.4	HQC	95.5	5.5	87.6	2.5
methionine sulfone	LQC	98.3	11.7	91.7	5.0	MQC	97.8	9.0	102.1	13.0	HQC	87.4	3.2	100.6	7.5
L-α-amino-n-butyric acid	LQC	104.0	9.3	98.7	10.9	MQC	104.7	10.6	111.9	9.4	HQC	108.3	3.9	95.8	7.6
sarcosine	LQC	101.4	6.8	112.4	7.4	MQC	99.7	7.3	100.1	12.5	HQC	94.8	5.1	100.6	7.6
β-alanine	LQC	103.9	10.4	101.9	11.7	MQC	100.5	8.2	108.6	9.8	HQC	99.4	6.7	112.2	10.7
L-alanine	LQC	103.1	5.8	101.3	8.1	MQC	102.9	5.5	98.9	7.1	HQC	98.7	5.5	89.9	4.0
trans−4-hydroxy-L-proline	LQC	98.0	6.3	103.2	5.9	MQC	101.3	7.2	96.2	6.4	HQC	100.0	5.0	104.2	7.8
L-threonine	LQC	99.9	8.9	97.0	13.0	MQC	106.7	7.1	105.1	10.4	HQC	101.7	6.2	99.9	3.9
L-glycine	LQC	92.6	8.9	98.7	9.9	MQC	90.2	5.7	99.9	5.4	HQC	93.9	4.6	90.6	4.5
L-glutamine	LQC	103.4	4.0	98.1	9.2	MQC	110.8	5.4	107.1	8.0	HQC	95.5	3.0	107.8	11.8
L-serine	LQC	99.1	7.2	101.5	6.8	MQC	100.6	5.5	99.9	9.2	HQC	97.2	5.3	97.9	6.8
L-asparagine	LQC	101.6	10.1	98.1	10.3	MQC	102.0	5.4	100.1	10.4	HQC	98.2	4.5	96.7	4.0

Table 4 (continued)

AAs		Intra-day		Inter-day			Intra-day		Inter-day			Intra-day		Inter-day	
		Accuracy (%)	Precision (RSD, %)	Accuracy (%)	Precision (RSD,%)		Accuracy (%)	Precision (RSD, %)	Accuracy (%)	Precision (RSD, %)		Accuracy (%)	Precision (RSD%)	Accuracy (%)	Precision (RSD, %)
L-glutamic acid	LQC	105.8	8.6	96.8	4.4	MQC	97.7	4.4	96.7	4.1	HQC	103.8	4.7	100.2	6.7
L-citrulline	LQC	106.9	10.5	97.8	13.5	MQC	105.0	9.5	109.6	7.6	HQC	103.5	5.5	112.8	5.7
1-methyl-L-histidine	LQC	102.8	5.1	99.7	8.0	MQC	104.8	6.9	104.8	8.8	HQC	101.9	3.8	94.8	6.8
L-aspartic acid	LQC	98.9	7.5	96.7	11.1	MQC	90.0	6.1	97.2	11.7	HQC	94.7	4.6	105.6	8.0
L-histidine	LQC	100.9	8.1	103.0	10.2	MQC	99.4	6.4	102.0	4.8	HQC	101.2	8.8	109.5	9.9
3-methyl-L-histidine	LQC	104.5	5.2	95.0	7.2	MQC	96.2	3.4	91.2	8.0	HQC	92.5	5.3	98.5	9.1
L-arginine	LQC	90.8	6.8	98.4	13.7	MQC	91.8	8.7	105.0	10.6	HQC	102.1	4.9	108.2	7.1
L-lysine	LQC	90.0	5.3	90.2	9.2	MQC	99.9	8.4	103.4	9.0	HQC	97.1	5.5	98.9	6.6
homocystine	LQC	102.5	4.9	86.5	8.8	MQC	99.2	9.3	89.3	6.6	HQC	95.1	3.9	93.6	8.0
L-ornithine	LQC	107.3	8.2	106.2	12.1	MQC	111.6	2.9	110.1	11.1	HQC	113.4	7.3	101.4	9.3
anserine	LQC	104.3	6.6	90.8	3.9	MQC	100.0	5.0	102.8	6.3	HQC	92.3	5.5	91.0	5.7
carnosine	LQC	102.2	8.1	90.3	8.1	MQC	100.4	9.0	103.6	8.7	HQC	89.0	6.3	87.1	9.5
δ-hydroxylysine	LQC	100.9	7.6	94.6	10.7	MQC	102.4	6.5	96.2	6.6	HQC	97.6	7.8	95.2	6.0
cystathionine	LQC	95.8	5.3	100.5	8.7	MQC	103.6	5.6	91.4	9.1	HQC	96.1	3.8	100.1	9.0
L-cystine	LQC	95.6	6.9	93.0	7.5	MQC	92.2	5.9	94.8	1.8	HQC	97.9	3.5	98.6	9.5

Table 5 Intra-day and Inter-day precision and accuracy of the analyzed AAs and their related compounds in stripped cyst fluids ($n=6$)

AAs		Intra-day		Inter-day			Intra-day		Inter-day			Intra-day		Inter-day	
		Accuracy (%)	Precision (RSD, %)	Accuracy (%)	Precision (RSD,%)		Accuracy (%)	Precision (RSD, %)	Accuracy (%)	Precision (RSD, %)		Accuracy (%)	Precision (RSD%)	Accuracy (%)	Precision (RSD, %)
N-acetyl-l-cysteine	LQC	96.5	11.1	86.5	3.1	MQC	101.3	5.6	91.0	7.2	HQC	100.5	6.7	93.6	13.8
3-cyclohexyl-D-alanine hydrate	LQC	105.9	7.6	113.1	7.4	MQC	94.3	11.2	96.4	10.8	HQC	111.2	7.5	108.4	7.4
3-nitro-L-tyrosine	LQC	92.1	5.0	110.6	12.5	MQC	98.8	8.4	108.7	10.6	HQC	104.0	7.0	112.2	9.6
L-phenylalanine	LQC	94.0	5.8	94.8	8.8	MQC	89.3	5.3	85.9	3.4	HQC	105.4	3.1	102.4	3.0
L-tryptophan	LQC	97.5	2.5	98.9	5.7	MQC	89.1	1.4	87.0	2.4	HQC	103.0	2.2	102.4	2.1
L-5-oxoproline	LQC	93.8	13.1	93.5	10.5	MQC	90.6	9.1	87.3	10.6	HQC	103.5	6.3	97.0	10.6
kynurenine	LQC	92.0	9.5	93.9	6.5	MQC	88.3	4.3	85.6	3.3	HQC	103.5	3.0	101.2	4.7
L-leucine	LQC	88.3	6.9	100.5	10.4	MQC	100.0	3.7	98.1	11.6	HQC	93.7	8.5	95.5	9.9
L-isoleucine	LQC	103.2	10.4	109.5	12.6	MQC	101.2	14.5	100.4	8.2	HQC	99.1	9.2	98.7	5.0
L-(+)-alpha-phenylglycine	LQC	98.4	10.7	93.9	6.7	MQC	99.1	14.4	101.5	13.6	HQC	100.3	6.8	93.6	9.4
ethanolamine	LQC	97.1	14.4	97.4	12.1	MQC	90.3	9.5	86.5	12.7	HQC	101.7	8.9	101.8	6.1
(R)-(-)−2-(2,5-dihydrophenyl) glycine	LQC	95.9	5.7	91.1	7.0	MQC	95.1	9.8	90.2	8.2	HQC	98.1	7.0	92.5	10.4
L-methionine	LQC	99.9	10.1	93.1	7.7	MQC	86.4	4.4	90.8	10.7	HQC	104.7	9.0	99.3	9.8
norvaline	LQC	109.6	12.3	105.0	11.9	MQC	87.2	5.5	87.6	8.4	HQC	101.1	8.2	99.9	8.4
γ-aminobutyric acid	LQC	97.5	4.6	99.5	8.5	MQC	96.5	5.9	91.8	9.6	HQC	104.5	6.8	99.9	9.8
L-valine	LQC	101.2	4.3	99.2	11.5	MQC	89.5	5.6	86.6	5.9	HQC	104.1	4.7	98.0	6.2
L-proline	LQC	100.7	5.1	97.6	5.5	MQC	90.5	5.1	89.2	3.3	HQC	101.8	2.4	97.3	4.1
DL-β-aminoisobutyric acid	LQC	96.4	9.0	99.7	9.3	MQC	91.6	6.2	86.4	2.1	HQC	103.4	8.0	91.2	8.3
L-tyrosine	LQC	97.3	6.6	96.7	5.1	MQC	89.9	4.4	85.4	6.7	HQC	104.4	5.1	102.9	5.4
L-taurine	LQC	95.9	12.6	87.2	9.9	MQC	94.3	9.4	88.4	8.2	HQC	96.3	5.8	90.1	9.8
L-cysteine	LQC	103.8	14.5	86.5	14.2	MQC	100.1	13.0	95.0	14.1	HQC	111.9	12.1	90.1	11.1
methionine sulfone	LQC	99.0	6.2	90.8	9.0	MQC	87.2	9.2	87.8	6.4	HQC	102.1	9.3	92.0	10.7
L-α-amino-n-butyric acid	LQC	97.7	8.4	99.4	6.4	MQC	87.4	7.0	89.5	9.0	HQC	99.7	10.8	95.1	3.1
sarcosine	LQC	98.5	10.6	98.7	14.1	MQC	93.8	8.6	104.2	8.9	HQC	103.2	4.4	95.9	4.0
β-alanine	LQC	94.2	14.1	97.7	14.8	MQC	88.9	13.4	93.6	9.8	HQC	103.7	7.7	103.5	9.1
L-alanine	LQC	99.7	7.6	95.1	9.0	MQC	90.8	7.1	89.0	12.0	HQC	106.8	6.0	97.7	8.0
trans−4-hydroxy-L-proline	LQC	98.3	8.8	105.6	10.7	MQC	91.3	8.4	102.8	9.3	HQC	96.8	5.2	95.9	6.3
L-threonine	LQC	103.4	8.6	97.6	10.6	MQC	90.8	7.4	93.9	9.4	HQC	103.1	4.9	97.9	1.7
L-glycine	LQC	110.0	10.5	105.5	12.2	MQC	88.0	9.2	88.8	14.5	HQC	100.4	7.9	100.3	11.6
L-glutamine	LQC	101.5	6.2	101.3	7.2	MQC	91.4	4.7	110.0	5.6	HQC	101.4	2.3	97.7	3.9
L-serine	LQC	97.8	11.4	101.9	10.1	MQC	88.6	11.3	90.4	6.1	HQC	104.4	6.9	96.3	4.3
L-asparagine	LQC	98.1	11.1	93.1	13.6	MQC	91.8	4.5	89.4	6.8	HQC	97.1	4.5	97.9	6.9
L-glutamic acid	LQC	98.3	7.8	107.8	8.9	MQC	94.5	7.2	87.1	7.7	HQC	106.8	4.0	103.8	9.6

Table 5 (continued)

AAs	Intra-day			Inter-day		Intra-day			Inter-day		Intra-day			Inter-day	
	Accuracy		Precision	Accuracy	Precision		Accuracy		Precision	Accuracy	Precision		Accuracy	Precision	
	(%)		(RSD, %)	(%)	(RSD,%)		(%)		(RSD, %)	(%)	(RSD, %)		(%)	(RSD%)	(%)	(RSD, %)
L-citrulline	LQC	105.3	9.5	107.5	14.4	MQC	99.1		5.3	97.5	5.7	HQC	104.2	4.0	108.7	6.4
1-methyl-L-histidine	LQC	104.6	14.2	91.0	14.1	MQC	94.9		11.2	90.8	13.8	HQC	100.0	7.9	89.1	8.9
L-aspartic acid	LQC	103.7	14.0	93.2	14.8	MQC	87.2		9.8	99.5	6.4	HQC	98.3	10.1	97.1	10.7
L-histidine	LQC	102.6	4.9	102.8	6.8	MQC	89.3		7.8	87.5	4.5	HQC	104.1	4.3	101.4	6.0
3-methyl-L-histidine	LQC	99.8	3.7	87.0	12.4	MQC	93.2		10.4	97.7	14.5	HQC	92.4	7.5	97.7	14.3
L-arginine	LQC	102.9	8.7	109.3	9.6	MQC	93.9		4.4	96.3	6.2	HQC	101.9	3.5	108.7	9.2
L-lysine	LQC	100.2	8.0	96.7	9.5	MQC	91.0		5.9	90.2	8.5	HQC	88.0	5.2	93.9	8.7
homocystine	LQC	102.9	10.5	95.3	10.7	MQC	89.2		7.3	86.6	9.2	HQC	101.3	5.9	90.5	6.3
L-ornithine	LQC	106.7	6.6	108.8	9.2	MQC	96.0		6.3	99.1	8.3	HQC	97.2	4.0	102.2	7.3
anserine	LQC	98.3	5.7	87.0	9.2	MQC	92.3		4.9	86.2	8.8	HQC	96.7	4.3	91.5	8.8
carnosine	LQC	98.6	8.3	89.4	8.2	MQC	92.5		6.0	104.2	8.9	HQC	99.0	6.2	95.0	9.0
δ-hydroxylysine	LQC	99.6	11.5	87.1	10.0	MQC	86.8		9.3	92.0	8.2	HQC	94.0	8.3	105.4	8.3
cystathionine	LQC	104.4	10.4	108.4	10.8	MQC	92.0		8.3	96.1	11.6	HQC	106.7	5.4	109.8	10.2
L-cystine	LQC	97.4	12.5	92.9	13.0	MQC	90.5		5.6	89.3	8.2	HQC	104.1	5.0	100.9	5.9

meet a tolerance of 15%, except for LQC samples, which are permitted a tolerance of 20%. Despite narrowing the concentration difference range to 15%, from Figures S3 and S4 and Fig. 2, most compounds in the stripped matrix still meet the requirements. It is further demonstrated in this research that stripping the matrix can result in a suitable and reliable alternative matrix.

More detailed comparisons were carried out between the stripped matrix method and the real matrix in three different matrices: urine, plasma, and cyst fluid, along with a control analysis incorporating water as a replacement matrix. Based on the calibration curve slope ratio between the stripped matrix and the real urine matrix, most target analytes fall within the acceptable range of 0.8–1.2, except for compounds 3-nitro-l-tyrosine, methionine sulfone, and DL-β-aminoisobutyric acid (see Figure S3). On the other hand, an examination of the corresponding concentration differences revealed that the compounds 3-nitro-l-tyrosine, kynurenine, DL-β-aminoisobutyric acid, L-taurine, methionine sulfone, and L-arginine exceeded the baseline standard of 15% variation. In contrast, when water was used as a surrogate matrix, about one-third of the target compounds failed to meet the slope ratio requirement, and nearly half of the compounds showed concentration deviations beyond the ± 15% range, indicating clear limitations of water as a surrogate matrix.

In the presence of activated carbon, the level of proteins and lipids in the matrix was reduced, especially phosphatidylcholine, lysophosphatidylcholine, and sphingomyelin [30]. The majority of MEs are attributed to these substances. Therefore, after being stripped away by the activated carbon, the suppressive effect on the analyte in the ionization source is reduced, and the resulting signal becomes higher compared with that in the real matrix. Isotope-labeled internal standards are commonly used for calibration because they possess similar physical and chemical properties to analytes and co-elute with them. Thus, both undergo equivalent matrix effects within the ionization source. This effectively counteracts the effects of matrix suppression or enhancement [31]. It follows that when calibrated with isotope-labeled internal standards, the results detected from the true matrix and the stripped matrix are consistent. On the other hand, when an isotope internal standard calibration is not conducted, the analyte signal in the stripped matrix becomes stronger than the one in the true matrix as a result of the reduced matrix composition. It is precisely for this reason that enhanced signals are detected when no internal standards have been used during calibration.

Based on the comparison of the slope ratios of calibration curves for two alternative matrices and the true matrices, approximately one-third of analytes are subject to matrix effects when water is used as an alternative matrix, consistent with previous literature reports [26]. This outcome can be attributed in large part to the differences between the

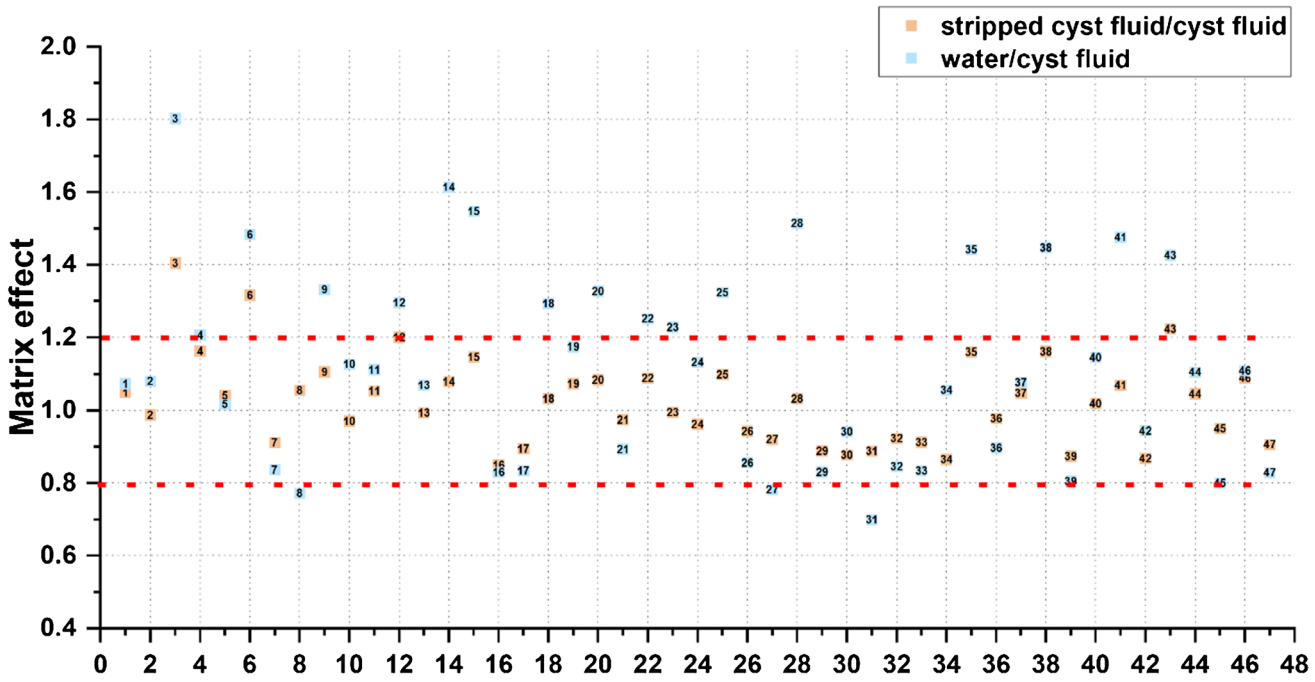


Fig. 1 Slope ratio between the surrogate matrix, water, and the real cyst fluid matrix. The x-axis represents the different analyte numbers, while the y-axis shows matrix effects (the ratio of the slopes obtained in the surrogate matrix (stripped matrix) and water, respectively, to those obtained in the authentic matrix). For each analyte, the respective slope ratios are indicated, with the analyte numbers fitted inside the symbols

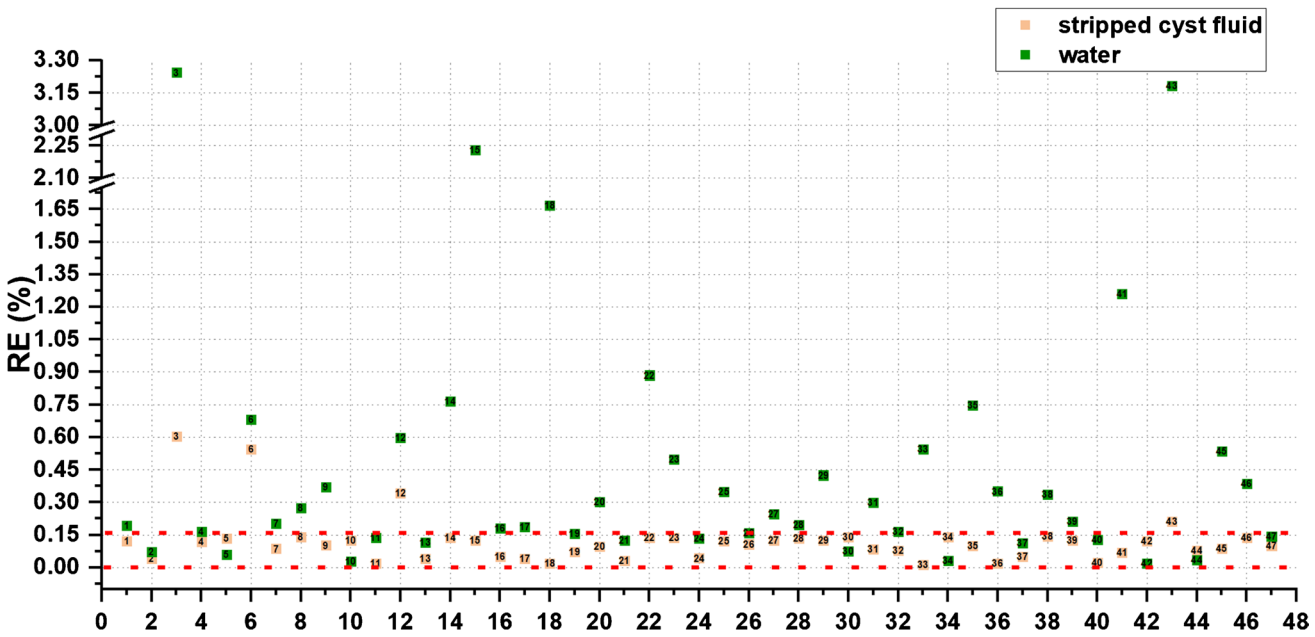


Fig. 2 Concentrations obtained using the standard addition method and the surrogate matrix method in cyst fluid. The x-axis represents the different analyte numbers, while the y-axis shows the relative error, defined as the difference between the analyte's concentration measured in the authentic matrix and that measured in the surrogate matrix (stripped matrix) and water, respectively, expressed as a percentage of the analyte's concentration measured in the authentic matrix (RE (%) = relative error). For each analyte, the respective RE values are indicated, with the analyte numbers fitted inside the symbols

composition of water and the actual substrate, particularly proteins, lipids, and salts, which are the factors that cause the MEs to occur. In contrast, the stripped matrix obtained after activated-carbon treatment more closely resembles the real matrix in composition, and therefore, the MEs it produces are also more comparable. It is possible to compensate for the suppression or enhancement of signals that may be caused by the matrix by using the corrective effect of isotope-labeled internal standards. As a result, the slope ratios between the stripped matrix and the real matrix for most analytes fall within the acceptable range of 0.8–1.2 for most analytes.

In the plasma matrix, the overall performance of the matrix-stripping method was relatively good. Only compounds L-cysteine and sarcosine had slope ratios outside the 0.8–1.2 range (see Figure S2), while compounds L-cysteine, methionine sulfone, sarcosine, and β-alanine did not meet the ± 15% concentration difference requirement. It can be suggested that the matrix stripping method effectively mitigates matrix effects in plasma. In contrast, if water is substituted for the matrix, a situation similar to that observed in urine is observed: nearly half of the compounds exhibit deviating slope ratios and concentration differences that are outside the acceptable range.

Cystic fluid was also assessed using the matrix stripping method, which similarly demonstrated high reliability, with only 3-nitro-l-tyrosine, l-5-oxoproline, and anserine having slope ratios that exceeded the acceptable range. Also, compounds 3-nitro-l-tyrosine, l-5-oxoproline, (R)-(-)−2-(2,5-dihydrophenyl)glycine, and anserine failed to meet the concentration difference criteria.

Stability

The stability of the analytes in urine was assessed under different conditions using LQC, MQC, and HQC samples in Table 6. After three freeze-thaw cycles, the stability ranged from 85.9% to 113.4%, with RSD values between 1.5% and 13.2%, which were within the generally accepted ± 15% criteria. Furthermore, the stability following storage at 4 °C for 1 day showed recovery values ranging from 85.7% to 113.7%, with RSD values between 2.1% and 13.6%. These stability tests indicated acceptable results, confirming that the method can produce reliable measurements under the tested storage and handling conditions.

The analytes exhibited good stability in cyst fluids under both refrigerated storage and freeze–thaw conditions (Table 7). At 4 °C storage, the stability remained within 87.1%–114.5% with RSDs of 1.8% –14.5%. After three freeze–thaw cycles, accuracy ranged from 85.2% to 114.6%, with RSDs between 1.1% and 14.7%. Stability assessments indicate that the method consistently delivers reliable quantitative results under the evaluated storage and processing conditions.

Recovery and dilution integrity

The recovery of AAs and related compounds in urine was evaluated using LQC, MQC, and HQC samples in Table 8. The recoveries were within 88.0%–113.3%, with RSD values ranging from 2.6% to 13.5%. The recovery values were within 85–115% and the RSD% less than 15%, which demonstrated good extraction efficiency and repeatability. For the dilution integrity test, the accuracy ranged from 86.6% to 111.8%, with RSD values between 3.4% and 10.6%, indicating that sample dilution did not affect the result obtained for urine samples.

Table 9 shows that in cyst fluid samples, recovery experiments conducted with LQC, MQC, and HQC samples yielded values within 85.5%–114.5%, with RSD values between 2.9% and 14.4%. These results fell well within the acceptance criteria for bioanalytical method validation, indicating that the extraction procedure provides consistent and efficient recovery of the analytes. The dilution integrity assessment showed accuracies within 89.6%–113.1% and RSD values of 2.4%–13.7%. These findings confirmed that diluting cyst fluid samples did not compromise quantification accuracy or precision, further verifying the method's suitability for analyzing cyst fluids samples.

Application of the LC-MS/MS assay

The urine samples of CKD and ADPKD patients were found to contain a total of 40 AAs and their related compounds. Figure 3 shows an analysis by orthogonal partial least squares discriminant analysis (OPLS-DA) of all the data. This revealed that CKD and ADPKD could be clearly distinguished in stages 1–2 and 3–4, indicating that there may be patterns of AA metabolism that differ between the two disease categories. Figure S5 shows the VIP score plot, which identified potential differential metabolites with VIP scores > 1 from urine samples at different stages of two kidney diseases. Urine sample results were corrected for creatinine. Figure S6 illustrates that three of these metabolites exhibited statistically significant differences ($p < 0.05$) upon further analysis of the kidney diseases. Figure S6 reveals that in the early stages (stages 1–2), citrulline levels were significantly higher in the ADPKD group than in the CKD group ($p < 0.05$). As the disease progressed, citrulline levels in the CKD group increased significantly [32]. There is evidence that citrulline may play a crucial role in the progression of chronic kidney disease, consistent with existing literature. At the same advanced stage, the levels of arginine ($p < 0.0001$) and threonine ($p < 0.05$) were higher in the ADPKD group than in the CKD group, with the findings

Table 6 Stability of the analyzed AAs and their related compounds in stripped urine ($n=6$)

AAs	Stability					Stability					Stability				
	Conc (μM)	4℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)	Conc (μM)	4 ℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)	Conc (μM)	4 ℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)
N-acetyl-l-cysteine	LQC	97.6	10.7	92.5	7.5	MQC	106.8	9.6	94.1	11.9	HQC	100.9	5.8	104.3	7.6
3-cyclohexyl-D-alanine hydrate	LQC	105.6	8.4	95.8	5.2	MQC	93.6	9.4	102.2	7.5	HQC	102.9	6.1	98.8	6.2
3-nitro-L-tyrosine	LQC	89.5	8.6	102.0	10.2	MQC	101.4	7.5	92.2	4.4	HQC	106.7	5.3	85.9	5.3
L-phenylalanine	LQC	87.1	9.9	99.6	9.7	MQC	92.5	4.5	106.5	3.7	HQC	98.2	3.2	100.0	4.0
L-tryptophan	LQC	96.7	8.5	95.8	10.5	MQC	100.9	6.1	99.5	11.3	HQC	102.6	5.5	100.9	5.0
L-5-oxoproline	LQC	100.4	10.0	90.2	12.0	MQC	109.0	8.6	102.7	10.5	HQC	111.5	6.3	98.0	6.3
kynurenine	LQC	89.6	9.9	100.9	10.9	MQC	89.2	4.8	112.4	9.6	HQC	106.0	5.5	101.3	6.9
L-leucine	LQC	97.3	11.8	102.0	11.0	MQC	102.2	10.4	96.6	10.8	HQC	104.4	5.8	98.6	9.1
L-isoleucine	LQC	97.8	6.0	104.3	8.4	MQC	97.9	4.0	92.8	4.9	HQC	106.3	5.6	105.8	2.1
L-(+)-alpha-phenylglycine	LQC	101.2	6.9	101.0	7.5	MQC	92.9	12.3	88.9	9.4	HQC	98.3	5.9	96.9	6.9
ethanolamine	LQC	105.2	4.1	98.8	8.0	MQC	102.7	7.8	86.8	5.0	HQC	105.2	4.1	101.3	6.7
(R)-(-)−2-(2,5-dihydrophenyl)glycine	LQC	90.3	9.1	97.2	9.9	MQC	94.6	10.9	95.5	6.8	HQC	96.8	8.3	101.9	6.7
L-methionine	LQC	104.0	7.9	98.4	10.7	MQC	100.8	6.4	100.8	4.4	HQC	102.7	5.3	93.9	5.2
norvaline	LQC	97.7	10.4	92.8	12.4	MQC	112.4	5.2	96.2	7.6	HQC	96.1	4.8	96.4	10.1
γ-aminobutyric acid	LQC	113.3	11.5	109.4	11.0	MQC	110.6	7.7	89.1	8.2	HQC	102.9	2.1	99.2	6.4
L-valine	LQC	108.8	8.0	98.0	4.0	MQC	91.2	8.4	107.8	9.9	HQC	97.5	5.7	104.2	6.9
L-proline	LQC	100.2	5.3	100.9	4.6	MQC	99.1	5.1	99.9	5.3	HQC	98.9	4.9	98.9	6.5
DL-β-aminoisobutyric acid	LQC	105.2	6.1	101.2	9.4	MQC	110.8	10.4	91.8	10.0	HQC	107.3	9.7	97.0	6.2
L-tyrosine	LQC	89.1	12.7	97.6	13.2	MQC	101.9	10.8	88.0	6.1	HQC	94.9	6.3	99.5	6.3
L-taurine	LQC	100.8	9.8	100.5	11.9	MQC	102.8	12.8	88.1	10.3	HQC	104.5	7.2	92.7	8.1
L-cysteine	LQC	101.8	13.6	96.8	7.4	MQC	101.2	9.6	97.7	5.5	HQC	96.1	3.7	98.5	4.2
methionine sulfone	LQC	102.2	11.2	98.2	7.5	MQC	103.7	8.5	103.4	4.5	HQC	109.4	8.1	108.1	7.3
L-α-amino-n-butyric acid	LQC	92.3	8.7	100.3	5.6	MQC	85.7	4.3	111.1	4.1	HQC	87.0	7.1	104.4	8.6
sarcosine	LQC	98.3	7.8	112.6	8.1	MQC	98.5	3.7	96.2	11.6	HQC	101.8	6.3	98.0	6.3
β-alanine	LQC	107.2	9.6	100.7	8.3	MQC	95.1	6.2	113.4	10.6	HQC	97.3	5.5	104.1	6.9
L-alanine	LQC	102.1	7.2	96.5	5.9	MQC	105.6	5.5	110.5	6.6	HQC	98.5	5.0	97.2	5.4
trans−4-hydroxy-L-proline	LQC	101.8	9.7	105.0	9.5	MQC	95.7	5.2	100.3	10.5	HQC	103.1	4.8	101.6	5.1
L-threonine	LQC	110.2	5.9	107.3	8.7	MQC	96.7	10.7	94.8	6.4	HQC	105.0	5.4	95.0	5.0
L-glycine	LQC	113.7	11.1	105.3	8.8	MQC	107.1	6.4	93.0	7.9	HQC	104.4	5.5	95.0	5.5
L-glutamine	LQC	94.8	6.8	100.6	9.1	MQC	92.7	8.1	104.5	5.4	HQC	87.4	6.7	110.9	1.5
L-serine	LQC	101.4	7.9	98.8	6.5	MQC	99.6	4.2	100.5	5.2	HQC	96.6	6.6	95.5	3.7
L-asparagine	LQC	95.6	7.4	98.2	8.9	MQC	102.7	7.8	100.1	8.0	HQC	100.7	6.2	100.5	6.7
L-glutamic acid	LQC	107.5	10.7	99.6	9.4	MQC	99.1	5.8	97.3	10.3	HQC	97.8	3.7	104.5	5.3
L-citrulline	LQC	96.5	9.5	106.8	6.8	MQC	95.1	6.1	112.6	8.0	HQC	92.0	5.8	105.5	5.9

Table 6 (continued)

AAs	Stability					Stability				Stability				
	Conc	4℃	RSD	Freeze-thaw	RSD	Conc	4 ℃	RSD	Freeze-thaw	RSD	Conc	4 ℃	RSD	Freeze-thaw	RSD
	(μM)	(%)	(%)	3 cycles (%)	(%)	(μM)	(%)	(%)	3 cycles (%)	(%)	(μM)	(%)	(%)	3 cycles (%)	(%)
1-methyl-L-histidine	LQC	104.6	6.5	106.1	6.3	MQC	99.1	7.7	113.2	5.7	HQC	94.5	7.8	100.9	7.0
L-aspartic acid	LQC	99.7	11.3	101.7	12.9	MQC	98.6	3.2	97.8	8.9	HQC	107.5	6.9	110.6	5.2
L-histidine	LQC	101.2	8.2	102.9	10.7	MQC	96.1	7.1	101.4	3.3	HQC	96.3	4.9	102.9	3.8
3-methyl-L-histidine	LQC	94.0	10.3	103.5	9.9	MQC	97.8	9.2	100.4	4.5	HQC	103.6	7.5	101.7	5.9
L-arginine	LQC	95.9	8.8	95.9	4.9	MQC	103.1	8.1	98.3	5.8	HQC	97.6	6.5	102.4	5.3
L-lysine	LQC	112.8	7.2	87.8	6.6	MQC	102.4	5.2	103.3	6.1	HQC	100.1	6.5	98.3	5.4
homocystine	LQC	102.9	11.5	90.6	9.0	MQC	88.1	8.7	91.7	7.5	HQC	98.4	5.2	90.0	6.5
L-ornithine	LQC	99.4	8.2	99.5	5.4	MQC	87.0	5.6	105.8	11.5	HQC	89.8	7.1	102.5	6.8
anserine	LQC	105.3	8.5	99.0	8.9	MQC	97.6	9.9	98.3	9.2	HQC	106.4	6.0	94.2	8.1
carnosine	LQC	90.4	7.3	93.4	8.9	MQC	94.1	8.4	100.6	10.5	HQC	95.2	5.0	106.4	4.1
δ-hydroxylysine	LQC	101.1	12.7	95.0	10.6	MQC	111.9	7.4	100.4	9.5	HQC	101.4	6.5	102.8	4.1
cystathionine	LQC	97.4	7.2	99.6	8.9	MQC	101.8	8.6	89.1	7.8	HQC	99.0	9.2	89.2	4.9
L-cystine	LQC	94.9	7.7	88.8	9.9	MQC	99.1	5.2	87.8	4.1	HQC	104.1	5.3	100.0	10.6

for arginine and threonine being related to what has been reported in the literature [33, 34]. It is generally observed that urinary excretion levels of most AAs decline with the progression of the disease. It is noteworthy that some AAs show similar reduction patterns in both CKD and ADPKD, suggesting that these AA decreases may be linked to the same pathological mechanisms behind renal impairments.

The plasma samples of this study contained a total of 36 AAs and related compounds. Figure 4 shows that OPLS-DA analysis at both stages 1–2 and 3–4 could separate the samples into two relatively distinct groups, indicating that ADPKD and CKD differ in their plasma AA patterns. Figure S7 presents the OPLS-DA analysis, which identified 11 potential differential metabolites with VIP scores greater than 1 from early-stage plasma samples of the two kidney diseases. Among them, three metabolites demonstrated statistically significant differences ($p < 0.05$) upon further evaluation in Figure S8. Applying the same approach to late-stage samples of the two kidney diseases revealed no significant statistical differences. Based on Figure S8, there was an increase in proline ($p < 0.0001$) and glutamic acid ($p < 0.001$) levels in the ADPKD group versus CKD during the early stages of the disease (stages 1 and 2). CKD patients have also been found to have lower proline and glutamic acid levels in previous literature reports [33]. The results of this study also revealed lower levels of sarcosine ($p < 0.001$) in the ADPKD group than in the CKD group. CKD patients have been reported to have increased plasma sarcosine concentrations [35]. The late stages 3–4 of the disease did not show any significant differences for the compounds studied. In part, this may be due to the limited sample size, which caused the statistical power to be reduced. The plasma AA levels of patients with both conditions generally decrease with the progression of the disease, which suggests that the decline in renal function affects the metabolism of AAs in general. It is worth noting that certain compounds, such as citrulline and 1-methyl-L-histidine, exhibited an upward trend. Several studies have demonstrated the relevance of these metabolites as biomarkers in the progression of kidney disease [36–38], and this finding is consistent with previous research.

As shown in Fig. 5, a total of 32 AAs and analogues were detected in the cystic fluid samples. Compared with plasma urine, the number of AA species was slightly lower, which may be related to the complexity of the matrix composition and the particularity of the metabolic environment. From an analytical perspective, because the LOQ for certain compounds was relatively a little higher in the cystic fluid, some of them did not reach the LOQ but were above LOD, such as sarcosine, cysteine, and hydroxylysine. Notably, some AAs have higher concentrations in cyst fluids than in plasma [39, 40]. Some substances associated with kidney disease, such as citrulline, proline, and kynurenine, increase

Table 7 Stability of the analyzed AAs and their related compounds in stripped cyst fluids ($n=6$)

AAs	Stability					Stability					Stability				
	Conc (μM)	4℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)	Conc (μM)	4 ℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)	Conc (μM)	4 ℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)
N-acetyl-l-cysteine	LQC	110.2	8.0	90.2	11.4	MQC	102.2	8.9	97.5	7.2	HQC	101.8	8.2	97.8	6.5
3-cyclohexyl-D-alanine hydrate	LQC	87.1	5.2	114.2	8.9	MQC	100.9	4.9	105.1	7.3	HQC	90.9	6.4	105.0	5.6
3-nitro-L-tyrosine	LQC	104.8	12.5	104.5	14.1	MQC	94.7	11.2	108.3	12.9	HQC	90.0	9.4	113.3	9.8
L-phenylalanine	LQC	106.7	5.8	99.7	8.3	MQC	112.2	5.5	88.9	5.5	HQC	94.9	3.1	104.5	2.8
L-tryptophan	LQC	100.1	6.4	97.6	5.8	MQC	113.0	2.9	90.3	4.5	HQC	95.9	2.6	107.2	2.2
L-5-oxoproline	LQC	108.5	13.7	101.7	13.5	MQC	112.8	4.4	87.7	10.9	HQC	90.3	5.5	101.4	9.1
kynurenine	LQC	105.6	5.4	94.2	6.6	MQC	113.4	4.2	92.2	6.2	HQC	95.6	2.6	106.9	5.7
L-leucine	LQC	109.4	13.9	95.0	12.7	MQC	109.6	12.3	92.4	8.3	HQC	113.0	8.7	96.4	6.4
L-isoleucine	LQC	98.0	10.8	109.8	13.2	MQC	108.1	4.4	106.0	12.3	HQC	89.8	6.1	97.5	4.7
L-(+)-alpha-phenylglycine	LQC	110.2	8.9	89.3	14.1	MQC	97.0	8.2	100.1	12.5	HQC	100.6	7.7	99.0	7.8
ethanolamine	LQC	108.6	11.5	95.2	9.3	MQC	112.1	9.2	88.0	8.1	HQC	101.9	8.4	105.8	9.9
(R)-(-)−2-(2,5-dihydrophenyl)glycine	LQC	113.6	7.8	90.0	14.5	MQC	102.9	10.9	95.4	13.6	HQC	100.1	8.6	97.9	8.4
L-methionine	LQC	96.2	5.1	97.0	8.3	MQC	101.8	10.7	90.1	6.7	HQC	99.8	7.6	105.4	9.1
norvaline	LQC	109.0	11.6	88.0	12.9	MQC	107.3	6.1	86.1	5.5	HQC	102.9	11.0	95.7	5.2
γ-aminobutyric acid	LQC	108.5	9.4	95.0	9.7	MQC	100.9	8.1	98.8	6.3	HQC	96.3	13.6	100.8	4.2
L-valine	LQC	99.4	6.9	87.0	12.5	MQC	113.9	4.0	90.6	6.1	HQC	95.0	2.2	94.1	4.9
L-proline	LQC	100.6	2.8	100.5	3.9	MQC	111.8	3.8	89.2	1.1	HQC	98.5	3.8	101.3	1.8
DL-β-aminoisobutyric acid	LQC	106.7	13.2	98.7	11.2	MQC	110.9	6.2	91.1	6.9	HQC	94.5	7.2	108.3	10.3
L-tyrosine	LQC	103.5	5.9	95.3	7.5	MQC	114.5	9.3	88.3	8.1	HQC	93.1	4.2	104.9	5.8
L-taurine	LQC	108.8	14.7	88.6	7.1	MQC	107.9	8.6	92.6	11.8	HQC	107.5	9.8	104.2	8.2
L-cysteine	LQC	110.9	12.0	87.6	11.6	MQC	100.2	9.5	101.3	11.8	HQC	91.4	7.1	100.2	13.2
methionine sulfone	LQC	99.6	5.9	103.4	12.5	MQC	112.8	4.3	85.9	4.3	HQC	98.1	12.9	104.9	4.5
L-α-amino-n-butyric acid	LQC	104.7	8.6	97.9	4.4	MQC	113.2	4.7	89.7	5.2	HQC	96.9	11.8	105.1	7.3
sarcosine	LQC	104.7	7.7	94.8	8.7	MQC	105.1	8.3	85.2	9.3	HQC	91.1	11.3	100.5	11.4
β-alanine	LQC	108.1	13.1	94.6	9.4	MQC	110.3	10.9	86.3	9.8	HQC	98.0	6.6	102.2	8.7
L-alanine	LQC	101.2	5.6	99.0	2.9	MQC	111.2	5.0	88.3	7.4	HQC	94.3	1.9	101.2	4.2
trans−4-hydroxy-L-proline	LQC	105.8	13.3	102.8	7.4	MQC	106.5	11.7	101.6	11.2	HQC	107.4	6.0	95.3	10.2
L-threonine	LQC	99.3	4.9	101.5	6.4	MQC	104.6	10.2	93.8	7.4	HQC	97.1	1.8	99.4	1.9
L-glycine	LQC	96.7	10.4	102.3	11.0	MQC	104.8	10.7	96.9	13.3	HQC	97.2	9.6	104.7	5.3
L-glutamine	LQC	101.3	7.2	99.9	3.4	MQC	110.0	5.6	93.2	3.5	HQC	97.7	3.9	101.3	5.4
L-serine	LQC	95.2	12.5	103.5	5.8	MQC	111.8	8.7	96.9	8.1	HQC	93.9	8.4	108.7	3.2
L-asparagine	LQC	102.5	5.3	97.0	8.5	MQC	108.0	3.9	92.7	7.1	HQC	99.6	5.1	102.1	10.8
L-glutamic acid	LQC	99.9	4.7	101.5	6.2	MQC	108.2	7.5	96.8	6.0	HQC	93.4	6.0	105.3	5.8
L-citrulline	LQC	97.7	7.3	106.4	12.9	MQC	108.1	11.6	96.7	13.0	HQC	97.9	5.0	108.7	9.6

Table 7 (continued)

AAs	Stability					Stability					Stability				
	Conc (μM)	4℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)	Conc (μM)	4 ℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)	Conc (μM)	4 ℃ (%)	RSD (%)	Freeze-thaw 3 cycles (%)	RSD (%)
1-methyl-L-histidine	LQC	100.8	11.1	102.3	13.0	MQC	113.0	5.9	87.3	8.8	HQC	101.7	6.4	100.9	7.9
L-aspartic acid	LQC	97.5	13.8	105.5	14.7	MQC	113.9	12.1	93.8	12.5	HQC	94.7	12.3	112.8	5.3
L-histidine	LQC	102.1	9.7	100.8	2.5	MQC	109.7	3.8	91.2	5.9	HQC	96.7	5.7	107.2	3.8
3-methyl-L-histidine	LQC	102.4	6.0	94.4	14.5	MQC	110.7	8.8	91.1	6.2	HQC	98.5	14.5	112.8	11.3
L-arginine	LQC	95.0	9.1	108.0	8.6	MQC	107.1	6.1	96.1	7.9	HQC	97.5	8.5	109.1	7.8
L-lysine	LQC	101.3	9.1	101.1	8.6	MQC	114.2	4.4	93.4	2.0	HQC	95.9	6.0	104.6	7.5
homocystine	LQC	94.7	11.5	104.3	6.9	MQC	113.8	7.6	90.5	6.9	HQC	96.5	8.3	102.8	10.1
L-ornithine	LQC	94.5	8.8	112.9	11.4	MQC	112.0	5.8	92.5	9.9	HQC	98.0	7.9	108.5	11.8
anserine	LQC	107.9	8.9	93.2	7.5	MQC	109.5	7.7	89.9	6.0	HQC	105.1	9.5	99.5	8.1
carnosine	LQC	103.9	6.0	96.3	2.5	MQC	113.8	9.3	89.3	5.8	HQC	101.6	10.6	100.9	7.7
δ-hydroxylysine	LQC	103.6	6.4	95.3	9.0	MQC	108.3	8.5	93.9	13.3	HQC	91.2	7.4	101.1	9.1
cystathionine	LQC	96.7	7.2	111.3	11.6	MQC	106.3	5.2	89.1	10.3	HQC	93.3	8.2	114.6	9.4
L-cystine	LQC	102.6	7.0	97.8	6.2	MQC	110.1	5.4	91.7	4.4	HQC	98.9	4.7	105.0	2.7

Table 8 Recovery and dilution integrity of the analyzed AAs and their related compounds in stripped urine ($n=6$)

AAs	Dilution integrity		Recovery			Recovery			Recovery	
	Accuracy (%)	Precision (%)	Recovery (%)	RSD (%)		Recovery (%)	RSD (%)		Recovery (%)	RSD (%)
N-acetyl-l-cysteine	93.6	6.2	LQC 101.7	6.7	MQC	99.6	8.3	HQC	95.4	7.4
3-cyclohexyl-D-alanine hydrate	94.4	10.5	LQC 92.5	6.4	MQC	101.1	9.4	HQC	100.6	10.3
3-nitro-L-tyrosine	99.8	4.8	LQC 96.5	10.7	MQC	92.1	6.4	HQC	97.4	10.4
L-phenylalanine	109.9	4.5	LQC 95.8	8.7	MQC	92.0	8.8	HQC	95.4	4.5
L-tryptophan	99.7	4.3	LQC 104.7	5.8	MQC	97.4	10.0	HQC	98.5	7.9
L-5-oxoproline	111.8	9.0	LQC 102.4	10.6	MQC	97.3	8.9	HQC	99.5	7.5
kynurenine	93.8	9.0	LQC 101.6	7.5	MQC	98.2	9.3	HQC	96.0	7.0
L-leucine	92.2	8.4	LQC 100.1	8.9	MQC	91.6	10.0	HQC	96.4	5.9
L-isoleucine	98.3	10.6	LQC 104.5	4.7	MQC	90.1	4.9	HQC	97.8	5.9
L-(+)-alpha-phenylglycine	99.9	6.8	LQC 103.3	8.2	MQC	93.3	11.1	HQC	91.6	8.0
ethanolamine	100.8	8.4	LQC 93.0	2.8	MQC	98.5	8.3	HQC	96.6	9.0
(R)-(-)−2-(2,5-dihydrophenyl)glycine	100.2	5.4	LQC 88.0	6.8	MQC	94.4	5.1	HQC	92.6	7.6
L-methionine	99.9	6.0	LQC 91.9	9.5	MQC	96.6	6.4	HQC	94.1	9.3
norvaline	93.9	9.7	LQC 101.1	6.9	MQC	100.4	7.4	HQC	113.3	4.3
γ-aminobutyric acid	95.2	8.5	LQC 95.7	6.1	MQC	106.3	5.2	HQC	97.0	4.1
L-valine	91.4	8.8	LQC 97.1	11.8	MQC	91.6	8.6	HQC	103.2	3.6
L-proline	98.1	3.6	LQC 111.9	9.6	MQC	103.5	7.0	HQC	94.6	7.7
DL-β-aminoisobutyric acid	94.7	7.9	LQC 101.6	7.7	MQC	103.0	7.3	HQC	101.5	6.5
L-tyrosine	102.5	5.6	LQC 98.3	13.0	MQC	102.0	11.3	HQC	106.1	7.6
L-taurine	105.2	8.8	LQC 110.4	3.9	MQC	102.9	8.9	HQC	96.5	6.9
L-cysteine	93.0	7.3	LQC 93.8	13.5	MQC	88.3	6.5	HQC	89.1	6.8
methionine sulfone	97.4	5.4	LQC 97.2	8.0	MQC	96.5	4.8	HQC	99.6	9.4
L-α-amino-n-butyric acid	101.7	3.4	LQC 102.3	10.6	MQC	109.1	3.5	HQC	95.0	5.5
sarcosine	101.9	5.4	LQC 91.4	9.0	MQC	96.4	12.9	HQC	98.7	9.3
β-alanine	111.6	5.7	LQC 88.6	8.6	MQC	99.6	5.6	HQC	105.7	7.4
L-alanine	103.4	4.3	LQC 92.3	6.5	MQC	101.6	9.3	HQC	93.7	12.4
trans−4-hydroxy-L-proline	98.5	10.5	LQC 97.6	9.4	MQC	93.3	7.6	HQC	107.3	8.6
L-threonine	86.6	7.8	LQC 97.5	9.0	MQC	89.7	8.4	HQC	92.3	8.5
L-glycine	96.5	9.1	LQC 108.7	6.5	MQC	102.0	9.6	HQC	91.2	7.9
L-glutamine	104.7	9.5	LQC 92.8	8.8	MQC	98.3	11.6	HQC	97.5	5.2
L-serine	100.0	4.5	LQC 98.7	9.3	MQC	89.8	10.0	HQC	98.5	10.4
L-asparagine	108.2	6.7	LQC 97.2	7.8	MQC	103.6	10.4	HQC	104.7	7.8
L-glutamic acid	103.7	3.4	LQC 96.6	5.4	MQC	97.7	11.0	HQC	96.1	6.2
L-citrulline	102.4	7.9	LQC 98.9	8.6	MQC	94.4	7.8	HQC	109.3	5.9
1-methyl-L-histidine	95.4	5.9	LQC 94.5	4.1	MQC	95.7	7.9	HQC	96.3	9.2
L-aspartic acid	103.0	6.5	LQC 109.7	5.9	MQC	107.1	4.5	HQC	102.9	6.3
L-histidine	94.3	9.2	LQC 96.3	7.6	MQC	91.8	5.4	HQC	93.7	6.9
3-methyl-L-histidine	95.3	6.1	LQC 107.1	5.2	MQC	105.5	8.0	HQC	100.4	6.7
L-arginine	92.5	7.6	LQC 88.7	2.7	MQC	101.3	7.8	HQC	112.5	5.0
L-lysine	108.2	7.8	LQC 105.6	8.1	MQC	111.6	6.1	HQC	96.8	7.3
homocystine	94.5	4.4	LQC 97.4	8.8	MQC	104.3	5.3	HQC	92.9	7.3
L-ornithine	92.0	5.0	LQC 100.3	2.6	MQC	98.7	13.2	HQC	89.6	9.7
anserine	102.3	6.6	LQC 93.1	7.4	MQC	96.1	7.6	HQC	89.7	8.3
carnosine	92.5	4.2	LQC 92.6	6.3	MQC	94.2	4.6	HQC	101.8	6.0
δ-hydroxylysine	90.7	5.1	LQC 93.5	7.5	MQC	90.6	8.2	HQC	98.9	2.9
cystathionine	100.6	6.8	LQC 97.8	6.1	MQC	100.6	9.7	HQC	97.2	3.2
L-cystine	101.6	8.0	LQC 96.9	9.9	MQC	93.8	10.4	HQC	107.8	9.1

Table 9 Recovery and dilution integrity of the analyzed AAs and their related compounds in stripped cyst fluids ($n=6$)

AAs	Dilution integrity			Recovery			Recovery			Recovery	
	Accuracy (%)	Precision (%)		Recovery (%)	RSD (%)		Recovery (%)	RSD (%)		Recovery (%)	RSD (%)
N-acetyl-l-cysteine	101.0	3.6	LQC	111.9	7.6	MQC	108.0	11.1	HQC	104.3	7.5
3-cyclohexyl-D-alanine hydrate	107.5	5.8	LQC	108.6	6.8	MQC	111.7	8.8	HQC	85.7	7.2
3-nitro-L-tyrosine	97.0	7.5	LQC	87.6	9.3	MQC	95.3	10.3	HQC	100.6	13.4
L-phenylalanine	101.4	3.6	LQC	97.2	4.3	MQC	114.5	4.9	HQC	102.0	3.0
L-tryptophan	105.4	2.5	LQC	96.4	6.3	MQC	110.9	4.1	HQC	99.4	3.4
L-5-oxoproline	90.5	3.9	LQC	102.2	14.1	MQC	99.2	11.1	HQC	111.4	11.7
kynurenine	105.4	3.0	LQC	95.6	3.9	MQC	108.2	6.0	HQC	94.1	3.6
L-leucine	90.4	8.6	LQC	105.3	8.5	MQC	92.0	11.8	HQC	90.4	6.2
L-isoleucine	98.3	7.6	LQC	102.6	9.8	MQC	88.7	12.2	HQC	93.5	8.7
L-(+)-alpha-phenylglycine	99.3	13.2	LQC	99.5	8.2	MQC	110.4	6.0	HQC	99.5	13.1
ethanolamine	98.8	4.2	LQC	113.8	10.2	MQC	102.1	12.7	HQC	95.9	7.1
(R)-(-)−2-(2,5-dihydrophenyl)glycine	97.3	13.7	LQC	104.7	10.6	MQC	109.6	3.6	HQC	94.6	10.9
L-methionine	99.6	8.9	LQC	93.3	10.5	MQC	107.6	12.0	HQC	102.4	7.9
norvaline	92.3	6.3	LQC	85.5	9.4	MQC	94.3	4.8	HQC	85.7	7.5
γ-aminobutyric acid	96.9	5.9	LQC	108.6	6.8	MQC	107.3	12.4	HQC	88.3	6.1
L-valine	97.3	3.9	LQC	90.5	5.3	MQC	97.7	5.1	HQC	112.8	8.6
L-proline	98.0	5.3	LQC	110.1	5.2	MQC	106.9	8.4	HQC	114.5	3.5
DL-β-aminoisobutyric acid	97.5	7.1	LQC	110.7	5.8	MQC	89.2	5.7	HQC	95.9	8.1
L-tyrosine	101.2	8.5	LQC	99.4	6.5	MQC	110.5	13.4	HQC	113.2	8.6
L-taurine	99.6	11.4	LQC	101.9	10.1	MQC	107.8	10.3	HQC	96.9	8.2
L-cysteine	113.1	9.3	LQC	111.0	12.9	MQC	112.9	6.3	HQC	112.5	9.3
methionine sulfone	90.9	7.0	LQC	109.6	7.9	MQC	109.4	3.8	HQC	97.0	8.2
L-α-amino-n-butyric acid	97.0	10.6	LQC	110.6	6.3	MQC	99.1	5.7	HQC	109.9	6.7
sarcosine	102.7	4.1	LQC	108.1	13.6	MQC	107.0	7.8	HQC	103.9	10.5
β-alanine	103.9	4.6	LQC	104.8	8.5	MQC	109.7	6.9	HQC	97.5	6.3
L-alanine	97.0	6.1	LQC	97.9	11.4	MQC	106.7	7.4	HQC	112.2	5.4
trans−4-hydroxy-L-proline	95.0	6.0	LQC	102.8	8.3	MQC	101.5	4.6	HQC	98.1	6.9
L-threonine	97.2	5.3	LQC	86.3	5.8	MQC	102.8	12.5	HQC	102.8	12.9
L-glycine	96.0	10.3	LQC	93.4	10.6	MQC	108.8	6.6	HQC	105.1	9.9
L-glutamine	98.1	2.4	LQC	99.5	8.0	MQC	99.9	6.0	HQC	106.3	6.6
L-serine	100.0	10.3	LQC	95.0	12.8	MQC	108.1	10.7	HQC	95.4	12.0
L-asparagine	94.1	5.4	LQC	102.9	13.5	MQC	112.6	5.9	HQC	99.5	8.8
L-glutamic acid	104.4	4.6	LQC	108.4	4.8	MQC	114.1	5.5	HQC	103.9	6.9
L-citrulline	101.9	3.5	LQC	108.6	11.0	MQC	110.0	5.9	HQC	102.3	10.2
1-methyl-L-histidine	89.6	11.8	LQC	97.1	8.7	MQC	100.6	14.1	HQC	107.9	7.2
L-aspartic acid	102.1	9.7	LQC	100.3	12.5	MQC	110.4	10.2	HQC	101.2	5.1
L-histidine	103.3	6.0	LQC	104.6	6.9	MQC	109.5	13.2	HQC	101.5	2.9
3-methyl-L-histidine	98.6	11.3	LQC	96.4	14.4	MQC	102.4	10.4	HQC	111.2	7.8
L-arginine	104.2	5.4	LQC	100.9	9.6	MQC	110.9	12.7	HQC	95.6	8.8
L-lysine	98.4	5.1	LQC	109.7	14.1	MQC	95.6	3.8	HQC	109.6	11.4
homocystine	92.8	4.4	LQC	94.7	5.1	MQC	99.4	8.8	HQC	96.9	9.9
L-ornithine	99.9	7.3	LQC	106.2	9.3	MQC	107.3	9.8	HQC	103.2	6.0
anserine	91.7	6.4	LQC	101.7	10.4	MQC	94.1	12.4	HQC	99.4	13.2
carnosine	93.2	10.3	LQC	100.5	9.5	MQC	94.1	8.2	HQC	96.4	9.1
δ-hydroxylysine	99.9	8.7	LQC	97.7	8.5	MQC	102.6	14.4	HQC	87.3	10.0
cystathionine	109.1	6.6	LQC	86.0	9.2	MQC	93.8	9.2	HQC	86.9	8.1
L-cystine	103.1	5.3	LQC	113.8	6.7	MQC	111.8	4.2	HQC	104.0	6.3

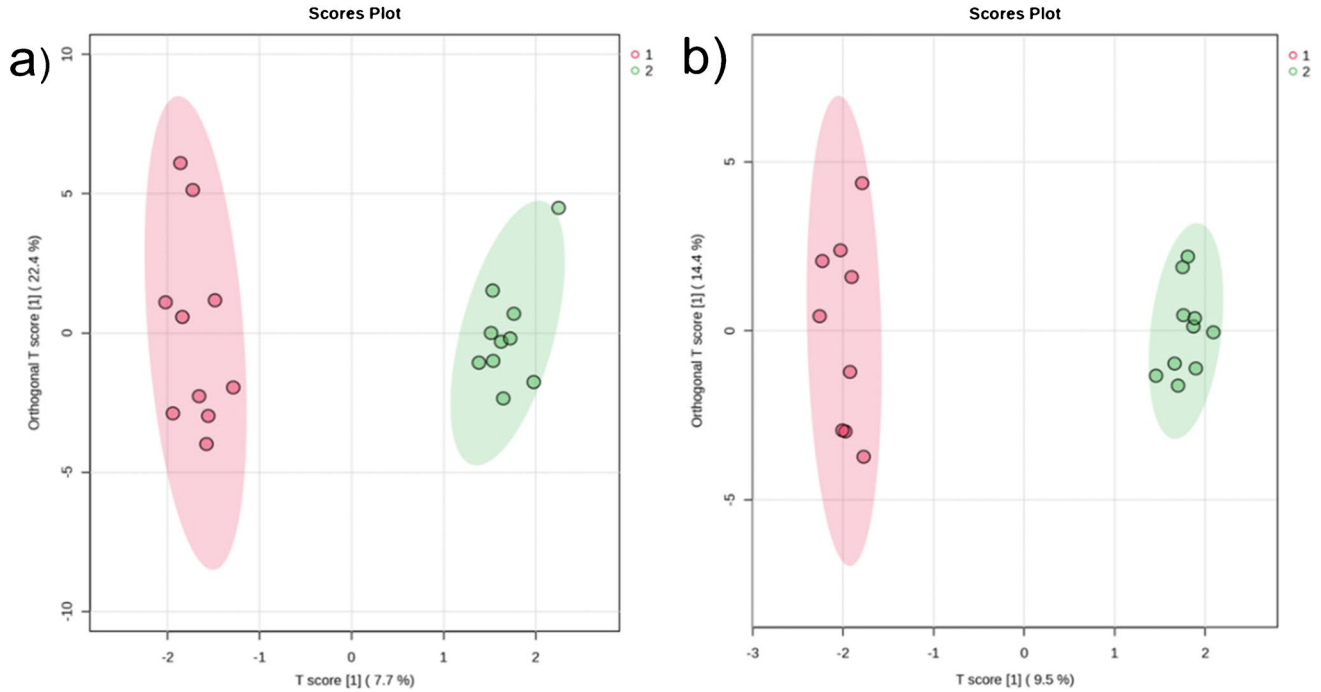


Fig. 3 OPLS-DA score plot in the kidney disease urine sample. **a** OPLS-DA score plot comparing CKD and ADPKD in early-stage (1–2) urine samples. **b** OPLS-DA score plot comparing CKD and ADPKD in late-stage (3–4) urine samples. CKD 1–2 ($n=10$), ADPKD1-2 ($n=10$); CKD1-2 ($n=9$), ADPKD ($n=10$). Red/left represents CKD, green/right represents ADPKD

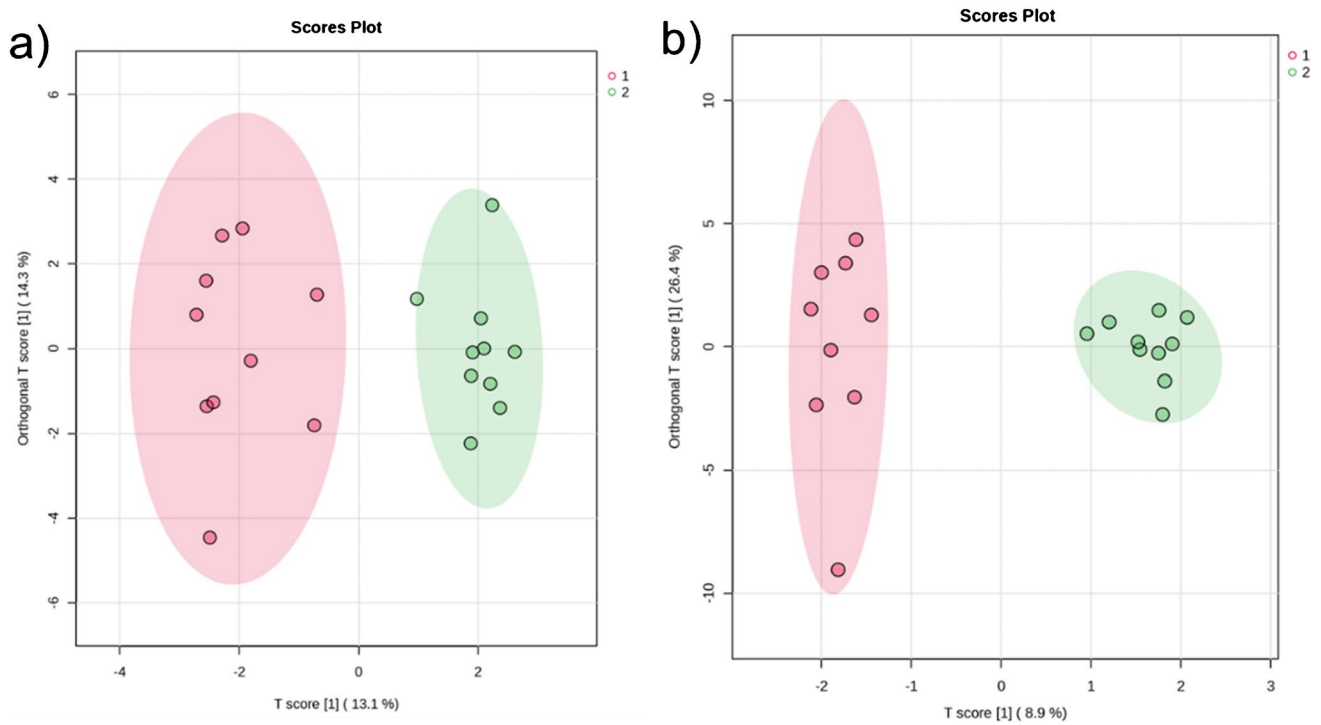


Fig. 4 OPLS-DA score plot in the kidney disease plasma sample. **a** OPLS-DA score plot comparing CKD and ADPKD in early-stage (1–2) plasma samples. **b** OPLS-DA score plot comparing CKD and ADPKD in late-stage (3–4) plasma samples. CKD 1–2 ($n=10$), ADPKD 1–2 ($n=10$); CKD 1–2 ($n=9$), ADPKD ($n=10$). Red/left represents CKD, green/right represents ADPKD

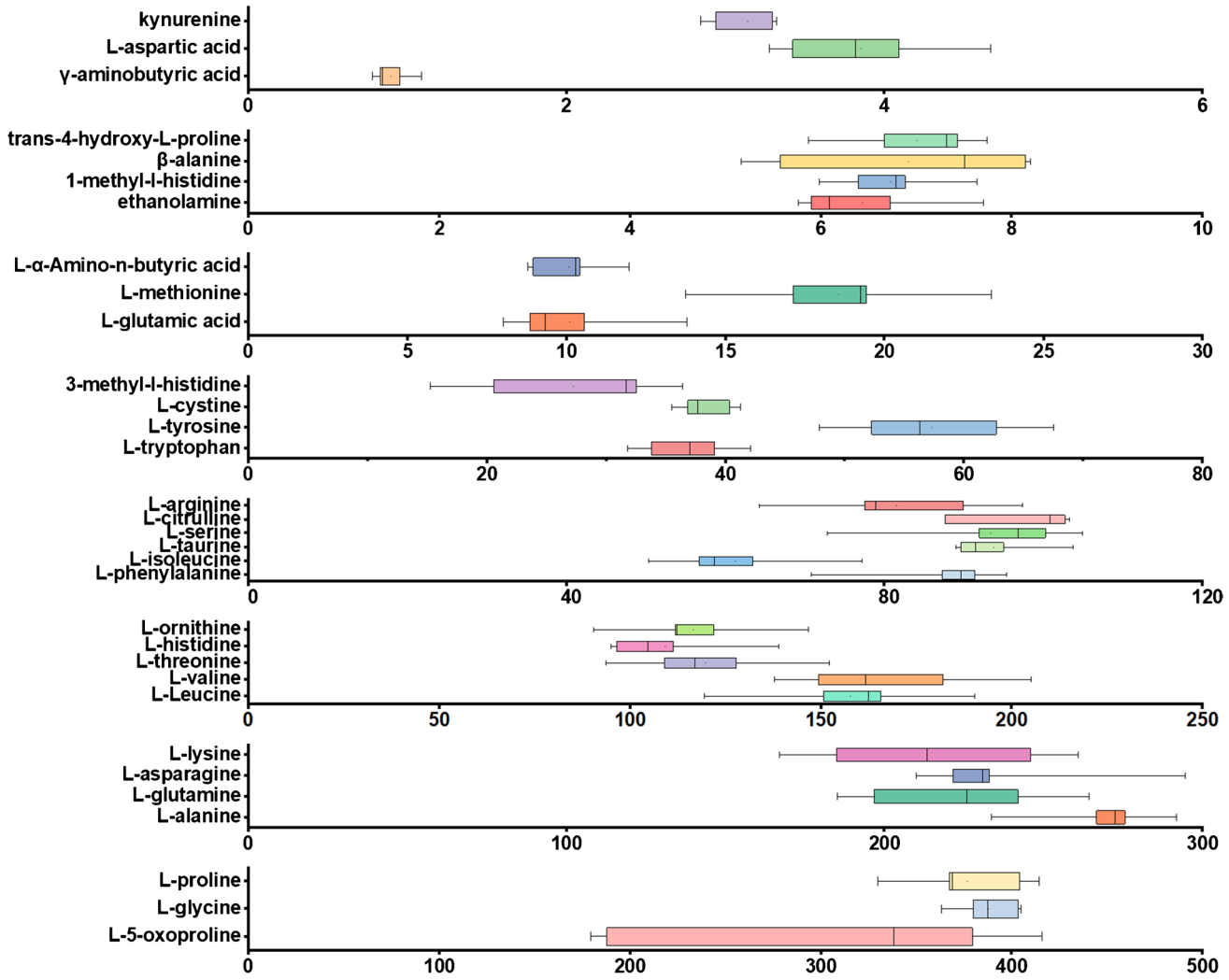


Fig. 5 Concentration of AAs and related compounds in the cyst fluids

in plasma as the disease progresses [37, 41], and their concentrations in cyst fluids are higher than the average plasma concentrations.

Conclusions

Biochemical samples contain many endogenous compounds. Blank matrices suitable for method validation are not readily available, making the construction of blank matrices for quantitative analysis difficult. The study seeks to overcome this limitation by preparing and stripping the matrix. Compared with pure solvents, the stripped matrix exhibits superior parallelism in the construction of calibration curves, which improves the reliability of quantitative results. By systematically validating the method, the study confirmed that the stripped matrix can be applied to the determination of AAs and further demonstrated its effectiveness. Three types of biological matrix samples have been investigated, providing robust technical support for investigations concerning the dynamic changes in endogenous AAs during the progression of kidney disease, with potential clinical implications.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00216-026-06687-9.

Acknowledgements Xiongwei Yin thanks the China Scholarship Council for financial support (grant 202108510041). C. Weydert is acknowledged for making available certain biological samples.

Author contributions Conceptualization: Ann Van Schepdael. Methodology: Xiongwei Yin. Formal analysis and investigation: Xiongwei Yin, Ann Van Schepdael. Data curation: Xiongwei Yin. Visualization: Xiongwei Yin. Writing—original draft preparation: Xiongwei Yin. Writing—review and editing: Jean-Paul Decuypere, Djalila Mekahli, Bert Bammens, Rudi Vennekens, Erwin Adams, Ann Van Schepdael.

Funding acquisition: Erwin Adams, Ann Van Schepdael. Project administration: Ann Van Schepdael. Resources: Jean-Paul Decuypere, Djalila Mekahli, Bert Bammens, Rudi Vennekens, Ann Van Schepdael. Supervision: Erwin Adams, Ann Van Schepdael.

Funding This study was funded by the China Scholarship Council.

Data availability The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

Declarations

Research involving human participants and/or animals The study involved human subjects and was approved by the Ethics Committee Research of UZ Leuven, Belgium, under number S50495.

Informed consent All subjects gave written, informed consent.

Conflict of interest The authors declare no competing interests.

Source of biological material UZ Leuven, Belgium.

References

1. Hu X, Guo F. Amino acid sensing in metabolic homeostasis and health. Endocr Rev. 2021;42(1):56–76.
2. Chrysopoulou M, Rinschen MM. Metabolic rewiring and communication: an integrative view of kidney proximal tubule function. Annu Rev Physiol. 2024;86:405–27.
3. Mitrofanova A, Merscher S, Fornoni A. Kidney lipid dysmetabolism and lipid droplet accumulation in chronic kidney disease. Nat Rev Nephrol. 2023;19(10):629–45.
4. Knol MGE, Wulfmeyer VC, Muller RU, Rinschen MM. Amino acid metabolism in kidney health and disease. Nat Rev Nephrol. 2024;20(12):771–88.
5. American Diabetes Association Professional Practice C. 11. Chronic kidney disease and risk management: standards of care in diabetes-2025. Diabetes Care. 2025;48(1 Suppl 1):S239-S251.
6. Boudonck KJ, Mitchell MW, Nemet L, Keresztes L, Nyska A, Shinar D, Rosenstock M. Discovery of metabolomics biomarkers for early detection of nephrotoxicity. Toxicol Pathol. 2009;37(3):280–92.
7. Naaman SC, Bakris GL. Diabetic nephropathy: update on pillars of therapy slowing progression. Diabetes Care. 2023;46(9):1574–86.
8. Baliga MM, Klawitter J, Christians U, Hopp K, Chonchol M, Gitomer BY, Cadnapaphornchai MA, Klawitter J. Metabolic profiling in children and young adults with autosomal dominant polycystic kidney disease. Sci Rep. 2021;11(1):6629.
9. Dahabiyeh LA, Nimer RM, Sumaily KM, Alabdaljabar MS, Jacob M, Sabi EM, Hussein MH, Rahman AA. Metabolomics profiling distinctively identified end-stage renal disease patients from chronic kidney disease patients. Sci Rep. 2023;13(1):6161.
10. Lee H, Jang HB, Yoo MG, Park SI, Lee HJ. Amino acid metabolites associated with chronic kidney disease: an eight-year follow-up Korean epidemiology study. Biomedicines. 2020. https://doi.org/10.3390/biomedicines8070222.
11. Hocher B, Adamski J. Metabolomics for clinical use and research in chronic kidney disease. Nat Rev Nephrol. 2017;13(5):269–84.
12. Ghafari N, Sleno L. Challenges and recent advances in quantitative mass spectrometry-based metabolomics. Anal Sci Adv. 2024;5(5–6): e2400007.
13. Danilova EY, Maslova AO, Stavrianidi AN, Nosyrev AE, Maltseva LD, Morozova OL. CKD urine metabolomics: modern concepts and approaches. Pathophysiology. 2023;30(4):443–66.
14. Raptis V, Kapoulas S, Grekas D. Role of asymmetrical dimethylarginine in the progression of renal disease. Nephrology (Carlton). 2013;18(1):11–21.
15. Maciel LS, Marengo A, Rubiolo P, Leito I, Herodes K. Derivatization-targeted analysis of amino compounds in plant extracts in neutral loss acquisition mode by liquid chromatography-tandem mass spectrometry. J Chromatogr A. 2021;1656: 462555.
16. Violi JP, Bishop DP, Padula MP, Westerhausen MT, Rodgers KJ. Acetonitrile adduct analysis of underivatised amino acids offers improved sensitivity for hydrophilic interaction liquid chromatography tandem mass-spectrometry. J Chromatogr A. 2021;1655: 462530.
17. van de Merbel NC. Quantitative determination of endogenous compounds in biological samples using chromatographic techniques. TrAC Trends Anal Chem. 2008;27(10):924–33.
18. Thakare R, Chhonker YS, Gautam N, Alamoudi JA, Alnouti Y. Quantitative analysis of endogenous compounds. J Pharm Biomed Anal. 2016;128:426–37.
19. Visconti G, Olesti E, Gonzalez-Ruiz V, Glauser G, Tonoli D, Lescuyer P, Vuilleumier N, Rudaz S. Internal calibration as an emerging approach for endogenous analyte quantification: application to steroids. Talanta. 2022;240:123149.
20. FDA. Center for drug evaluation and research, bioanalytical method validation guidance for industry. 2018. https://www.fda.gov/media/70858/download. Accessed 3 Aug 2026.
21. Mishra S, Rajput N, Jadav T, Sahu AK, Tekade RK, Sengupta P. Advancement in analytical strategies for quantification of biomarkers with a special emphasis on surrogate approaches. Crit Rev Anal Chem. 2023;53(7):1515–30.
22. Nelis M, Decraecker L, Boeckxstaens G, Augustijns P, Cabooter D. Development of a HILIC-MS/MS method for the quantification of histamine and its main metabolites in human urine samples. Talanta. 2020;220: 121328.
23. Voegel CD, Baumgartner MR, Kraemer T, Wust S, Binz TM. Simultaneous quantification of steroid hormones and endocannabinoids (ECs) in human hair using an automated supported liquid extraction (SLE) and LC-MS/MS - insights into EC baseline values and correlation to steroid concentrations. Talanta. 2021;222: 121499.
24. Su M, Drotleff B, Janker T, Burger Z, Kimmig AS, Derntl B, Lammerhofer M. Quantification of endogenous steroids and hormonal contraceptives in human plasma via surrogate calibration and UHPLC-MS/MS. Anal Chem. 2025;97(25):13496–503.
25. Wang XN, Liu JQ, Shi ZQ, Sun FY, Liu LF, Xin GZ. Orthogonal label and label-free dual pretreatment for targeted profiling of neurotransmitters in enteric nervous system. Anal Chim Acta. 2020;1139:68–78.
26. Virgiliou C, Theodoridis G, Wilson ID, Gika HG. Quantification of endogenous aminoacids and aminoacid derivatives in urine by hydrophilic interaction liquid chromatography tandem mass spectrometry. J Chromatogr A. 2021;1642: 462005.
27. Yin X, Baldoni I, Adams E, Van Schepdael A. Analysis profiling of 48 endogenous amino acids and related compounds in human plasma using hydrophilic interaction liquid chromatography-tandem mass spectrometry. Molecules. 2024. https://doi.org/10.3390/molecules29245993.
28. European Medicines Agency. Guideline on bioanalytical method validation, committee for medicinal products for human use; (EMEA/CHMP/EWP/192217/2009). Amsterdam, The Netherlands: European Medicines Agency; 2011.

29. Khamis MM, Adamko DJ, El-Aneed A. Strategies and challenges in method development and validation for the absolute quantification of endogenous biomarker metabolites using liquid chromatography-tandem mass spectrometry. Mass Spectrom Rev. 2021;40(1):31–52.
30. Uher M, Mičuda S, Kacerovský M, Hroch M. An alternative approach to validation of liquid chromatography-mass spectrometry methods for the quantification of endogenous compounds. J Chromatogr A. 2023;1705: 464173.
31. Zhou W, Yang S, Wang PG. Matrix effects and application of matrix effect factor. Bioanalysis. 2017;9(23):1839-44.
32. Duranton F, Lundin U, Gayrard N, Mischak H, Aparicio M, Mourad G, Daures JP, Weinberger KM, Argiles A. Plasma and urinary amino acid metabolomic profiling in patients with different levels of kidney function. Clin J Am Soc Nephrol. 2014;9(1):37–45.
33. Lin IC, Hsu CN, Lo MH, Chien SJ, Tain YL. Low urinary citrulline/arginine ratio associated with blood pressure abnormalities and arterial stiffness in childhood chronic kidney disease. J Am Soc Hypertens. 2016;10(2):115–23.
34. Gil RB, Ortiz A, Sanchez-Nino MD, Markoska K, Schepers E, Vanholder R, Glorieux G, Schmitt-Kopplin P, Heinzmann SS. Increased urinary osmolyte excretion indicates chronic kidney disease severity and progression rate. Nephrol Dial Transplant. 2018;33(12):2156–64.
35. Gervasini G, Verde Z, Gonzalez LM, Chicharro C, Gonzalez-Rodriguez L, Fernandez-Araque A, Mota-Zamorano S, Cancho B, Perez-Hernandez A, Garcia-Lopez V, Bandres F, Robles NR. Prognostic significance of amino acid and biogenic amines profiling in chronic kidney disease. Biomedicines. 2023. https://doi.org/10.3390/biomedicines11102775.
36. Yamaguchi Y, Zampino M, Moaddel R, Chen TK, Tian Q, Ferrucci L, Semba RD. Plasma metabolites associated with chronic kidney disease and renal function in adults from the Baltimore Longitudinal Study of Aging. Metabolomics. 2021;17(1):9.
37. Arefin S, Lofgren L, Stenvinkel P, Granqvist AB, Kublickiene K. Associations of biopterins and ADMA with vascular function in peripheral microcirculation from patients with chronic kidney disease. Int J Mol Sci. 2023. https://doi.org/10.3390/ijms24065582.
38. Guo X, Peng H, Liu P, Tang L, Fang J, Aoieong C, Tou T, Tsai T, Liu X. Metabolic profiling identifies the significance of caffeine metabolism in CKD. Front Bioeng Biotechnol. 2023;11:1006246.
39. Heckscher S, Ihlo NA, Schueler J, Kellermeier F, Werner JM, Nuebel B, Gross V, May M, Wullich B, Kammerl M. Metabolomic profiling of renal cyst fluid in advanced ADPKD: insights from dialysis and transplantation cohorts. Metabolomics. 2025;21(4):90.
40. Klemens CA, Fedoriuk M, Semenikhina M, Stefanenko M, Zietara A, Levchenko V, Dissanayake LV, Palygin O, Staruschenko A. Electrolyte and metabolite composition of cystic fluid from a rat model of ARPKD. Commun Biol. 2025;8(1):230.
41. Klawitter J, Jackson MJ, Smith PH, Hopp K, Chonchol M, Gitomer BY, Cadnapaphornchai MA, Christians U, Klawitter J. Kynurenines in polycystic kidney disease. J Nephrol. 2023;36(1):83–91.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.